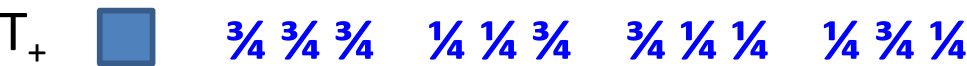
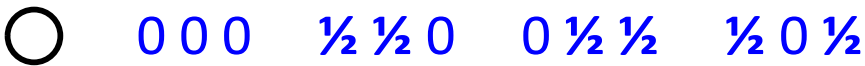
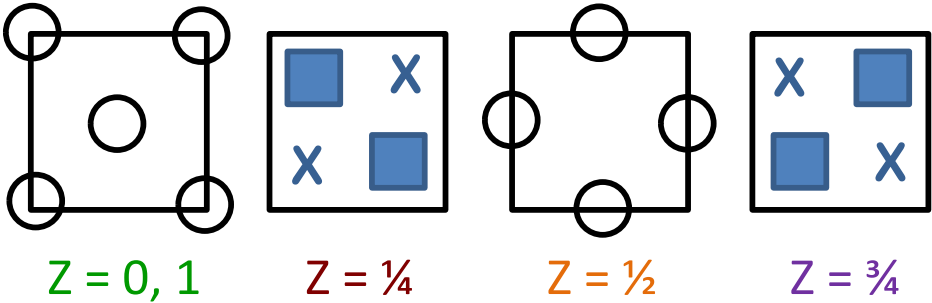
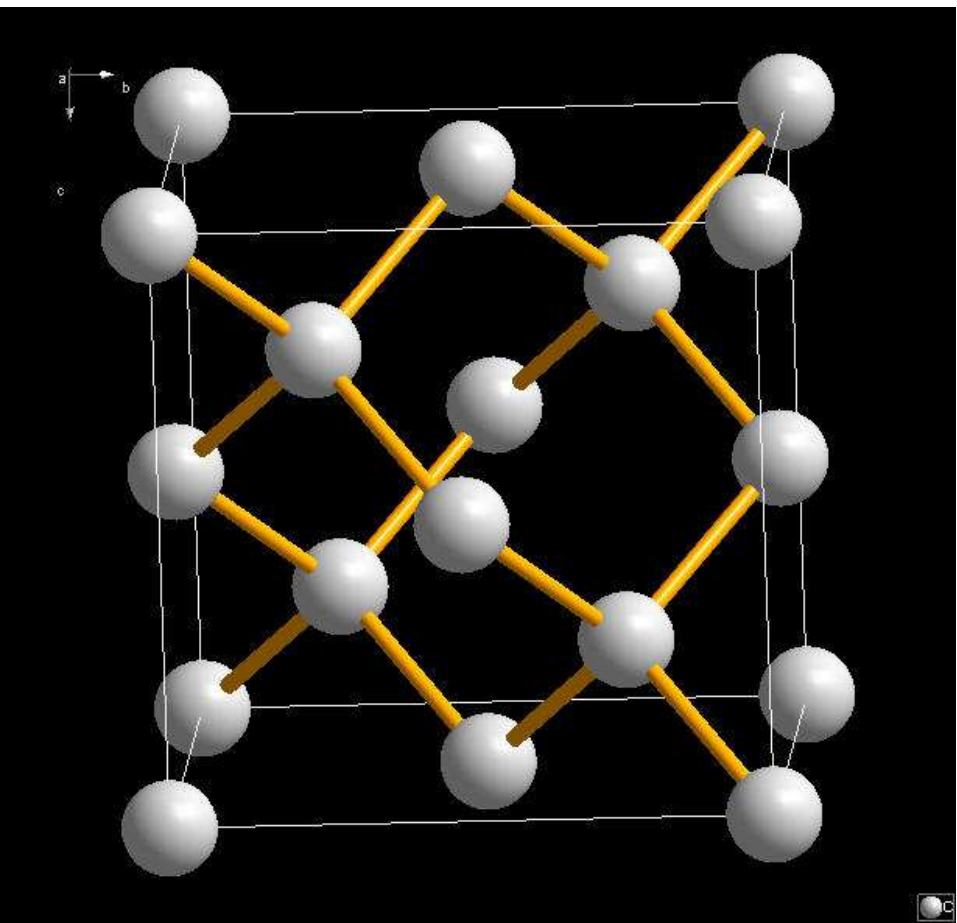


Special structures

Diamond



Covalent type structure

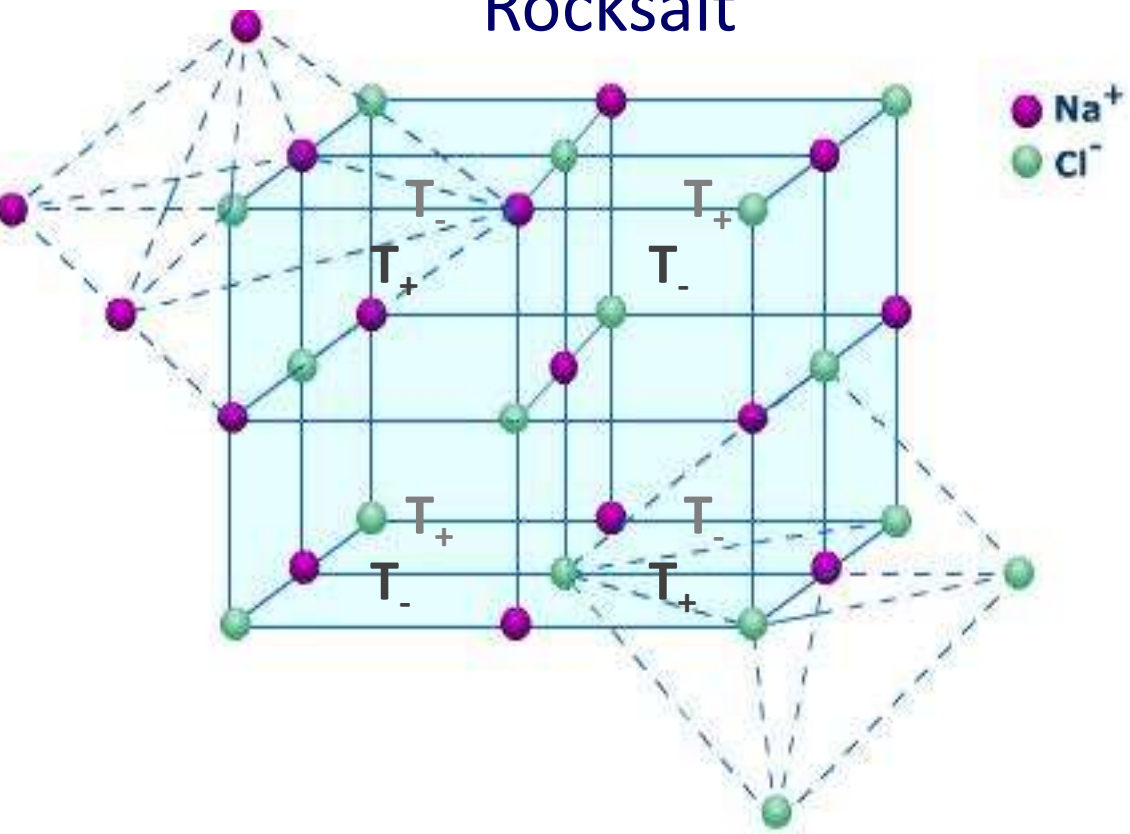
Alternate td voids are occupied by carbon atoms

Normal FCC structure: $\sqrt{2} a = 4 r$
 $\therefore a = \sqrt{8} r$
 $r = \text{radius of one atom}$

Diamond structure: $\sqrt{3} a = 8 r$
 $\therefore a = \sqrt{\frac{8}{3}} \times \sqrt{8} r$ $a \text{ increases by } 1.633 \text{ \AA}$

Rocksalt or Zinc blende or Antifluorite

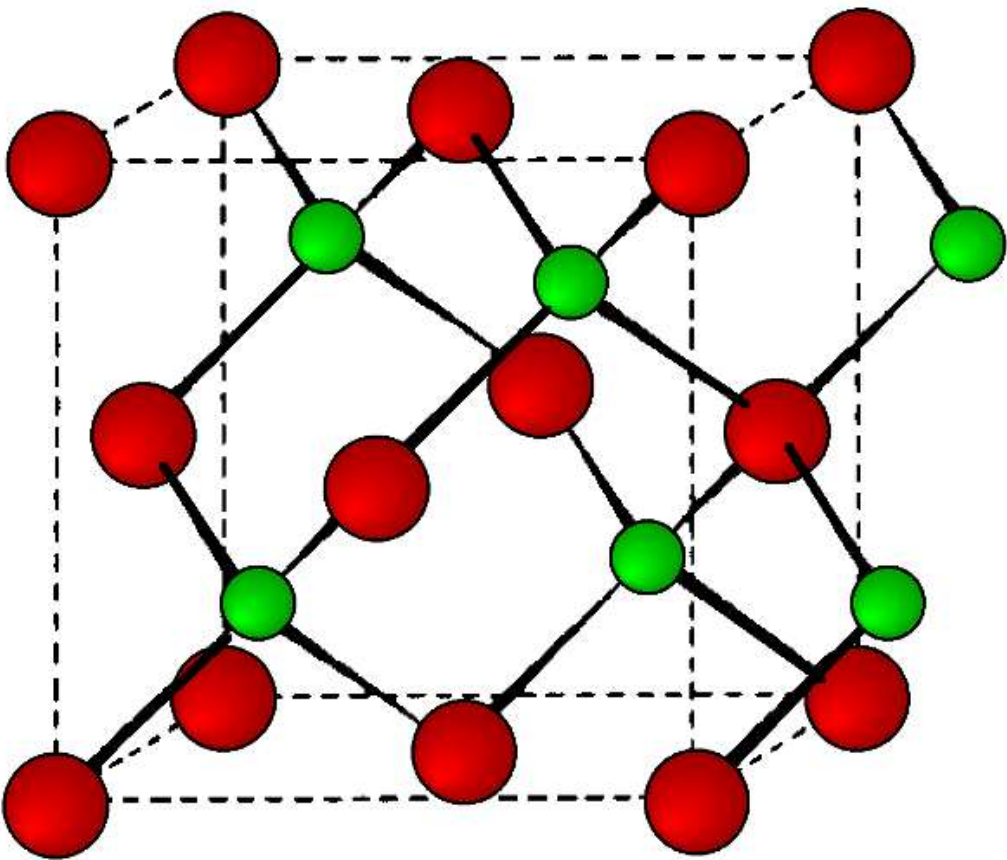
Rocksalt



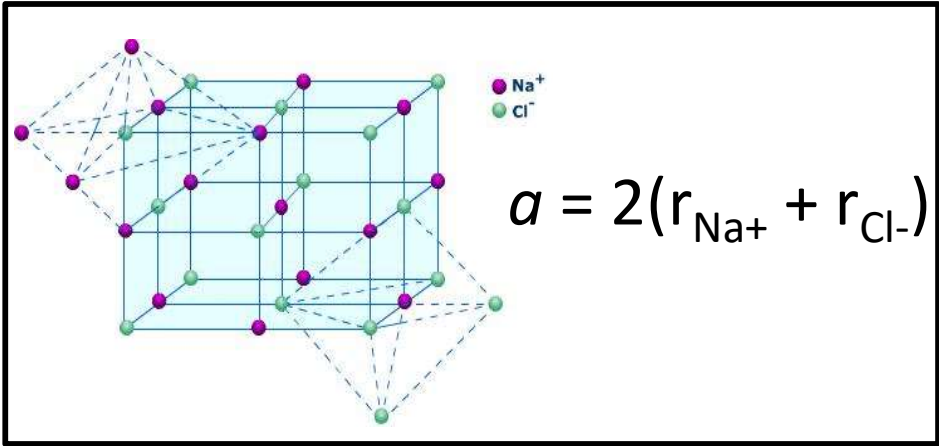
Oh	$\frac{1}{2}$	0	0	0	$\frac{1}{2}$	0	0	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
T ₊	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{1}{4}$
T ₋	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{4}$

- Rocksalt** (NaCl type) : Oh sites occupied by cations, T₊, T₋ empty
- Zinc blende** (ZnS type) : T₊ (or T₋) occupied, Oh, T₋ (or T₊) empty
- Antifluorite** (Na₂O type) : T₊ & T₋ occupied, Oh empty

Zinc Blende



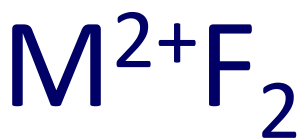
$$\frac{\sqrt{3}a}{2} = 2r_{S^{2-}} + 2r_{Zn^{2+}}$$



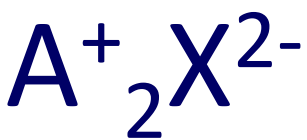
Zinc blende (ZnS type) : T₊ (or T₋) occupied, Oh, T₋ (or T₊) empty

CuS & CdS crystallize in this structure B3 type

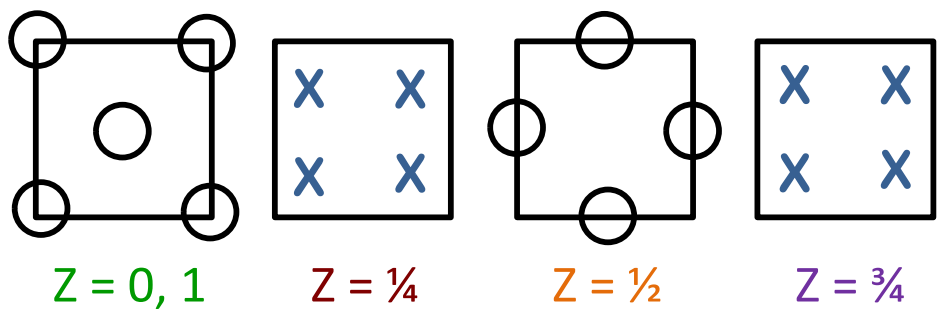
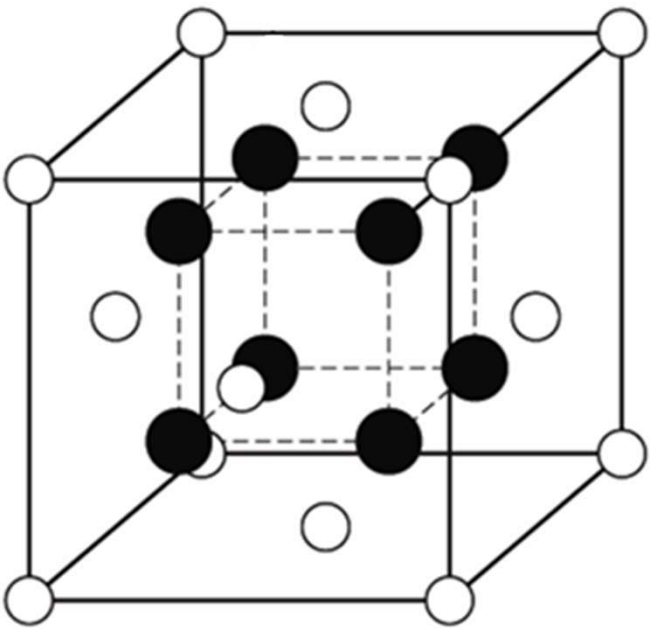
Fluorite & Antifluorite



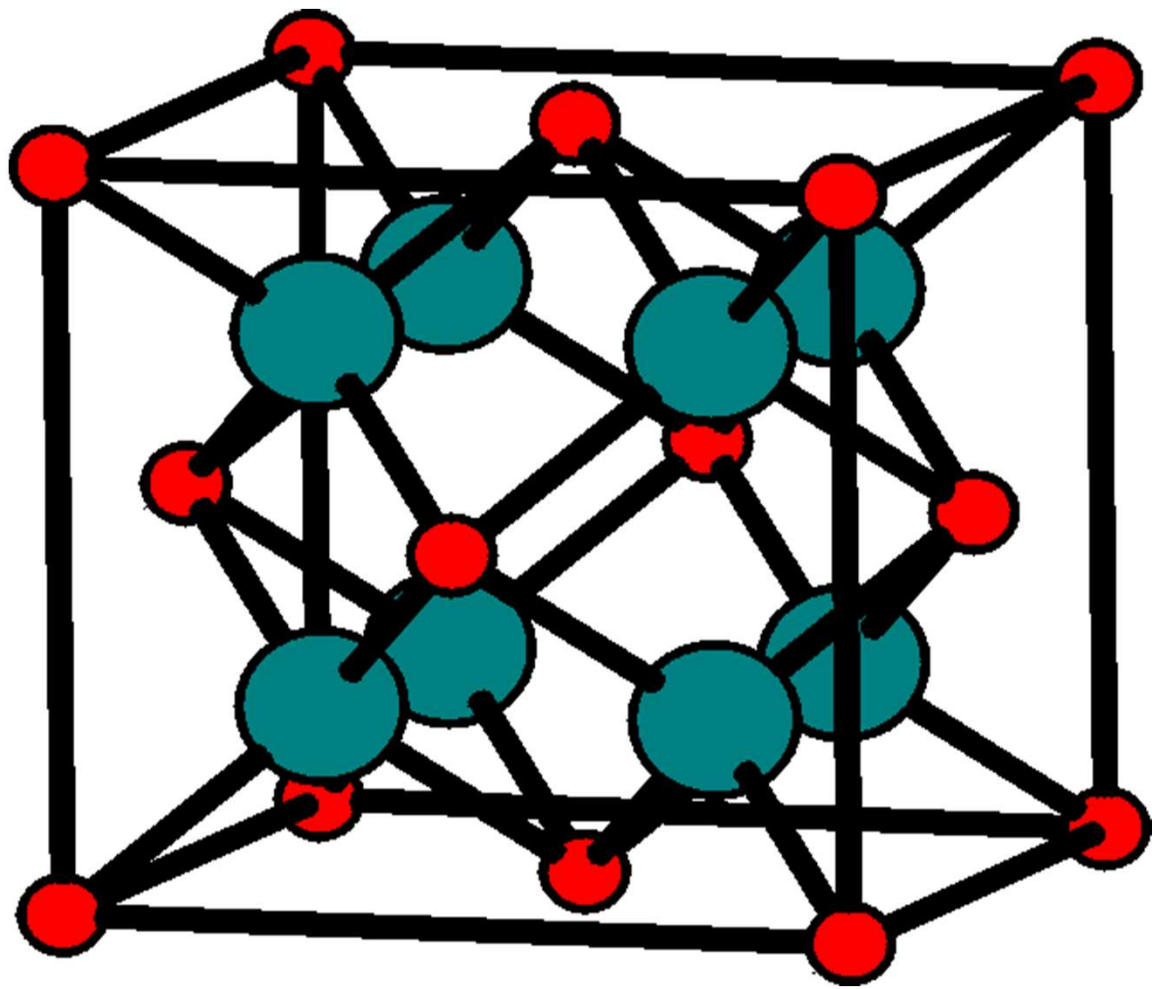
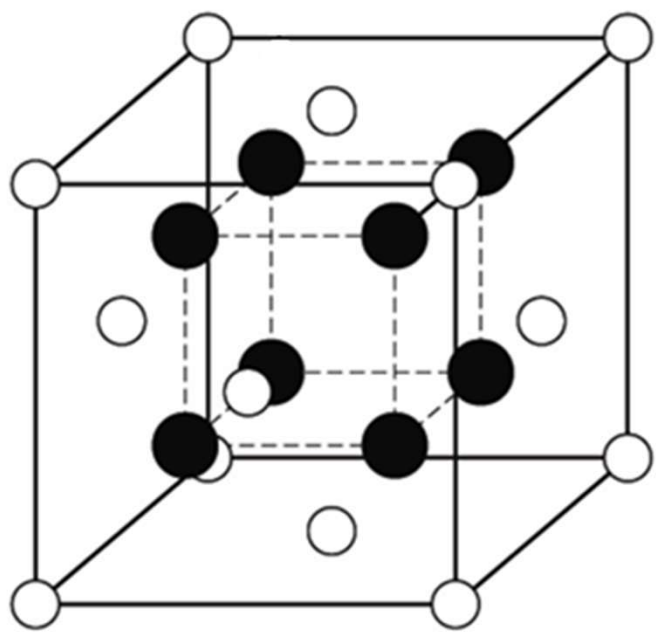
- Cations → 8 coordinate
- x Anions → 4 coordinate



- x Cations → 4 coordinate
- Anions → 8 coordinate

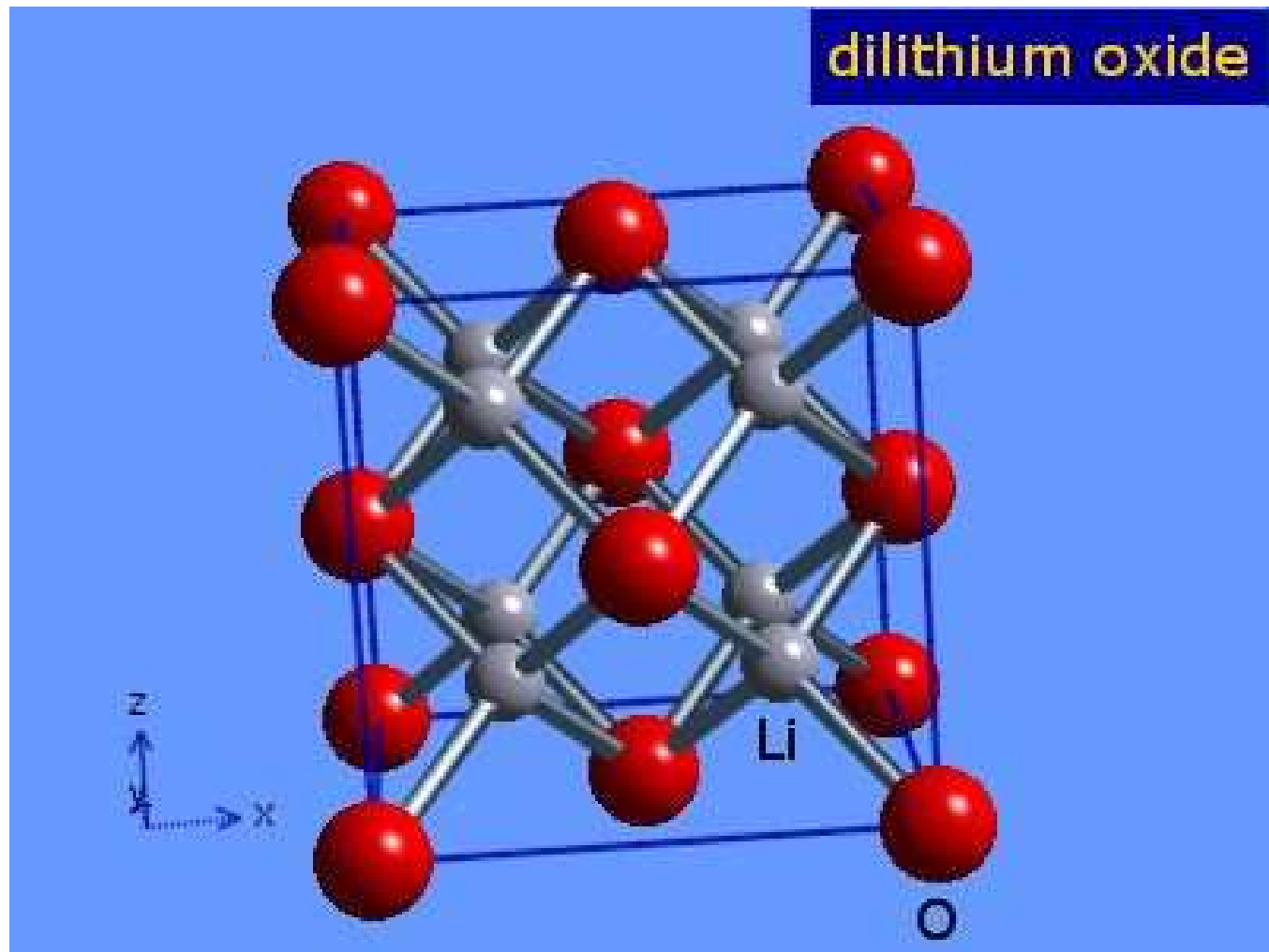
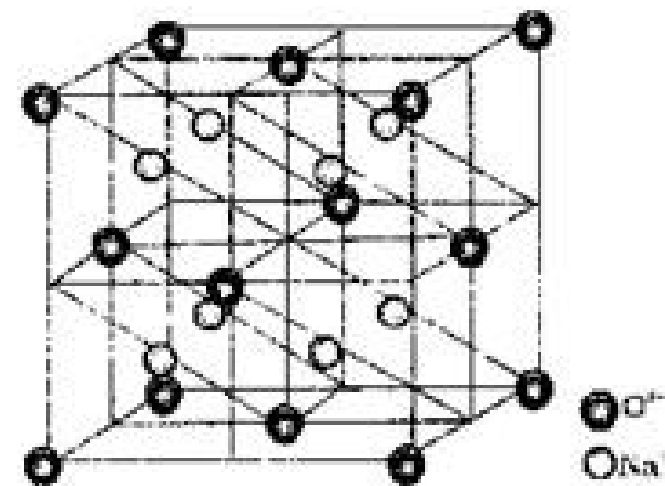


Fluorite



CaF_2 , SrF_2 , BaF_2 , SrCl_2 , BaCl_2 , CdF_2 , HgF_2 ,

Antifluorite

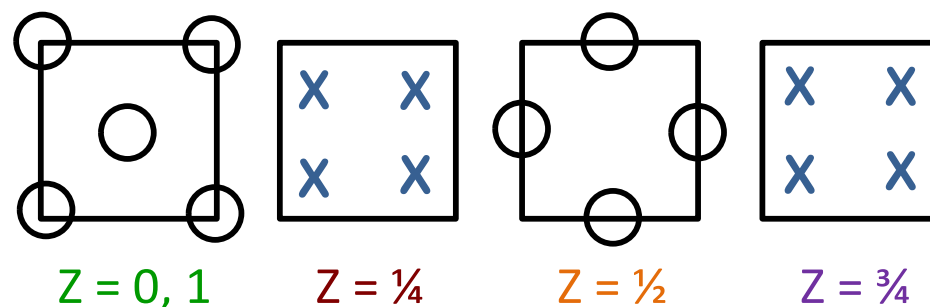
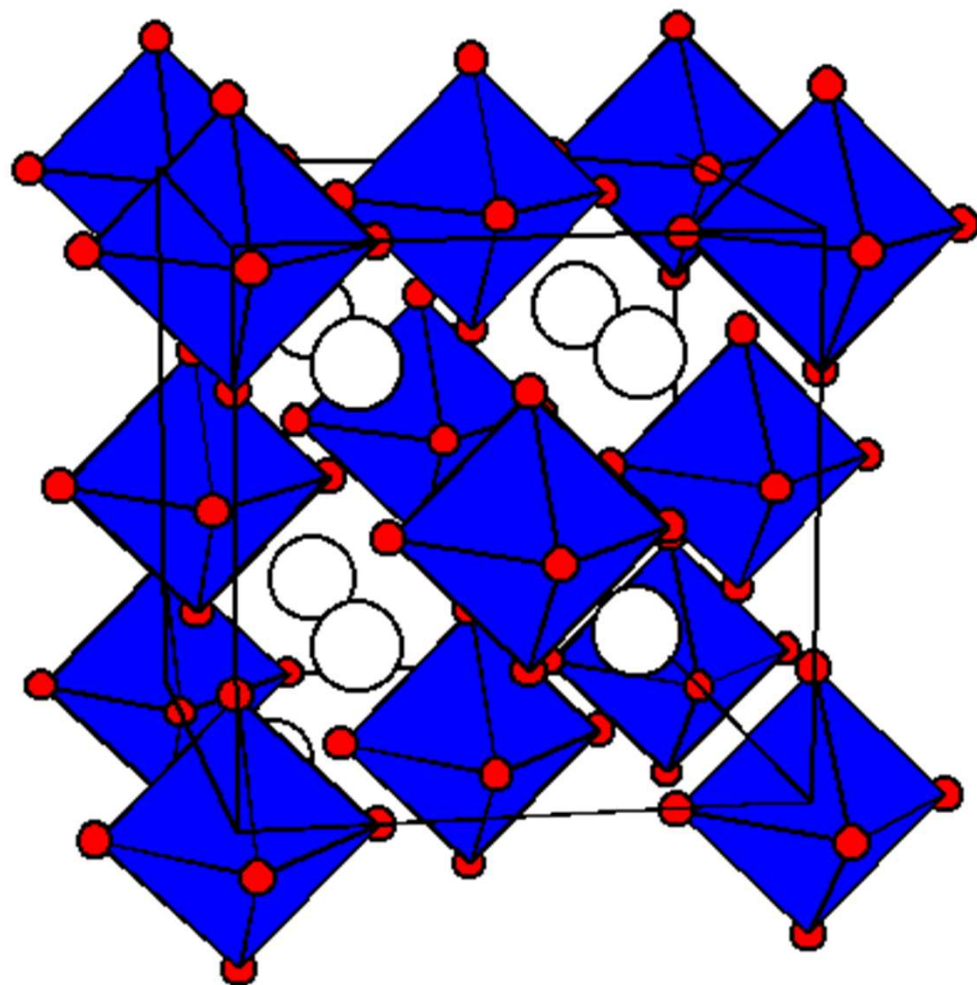


Li_2O , Li_2S , Li_2Se , Li_2Te , Na_2O , Na_2S , Na_2Se

Complicated Antifluorite Structure



A_2BX_6 type

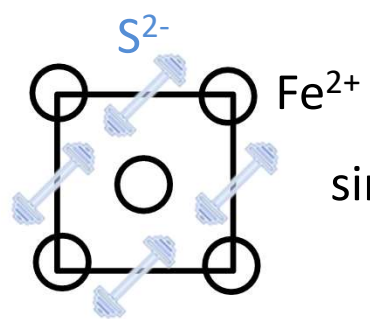
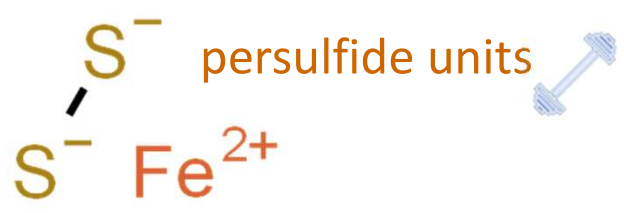


$\times$ K^+

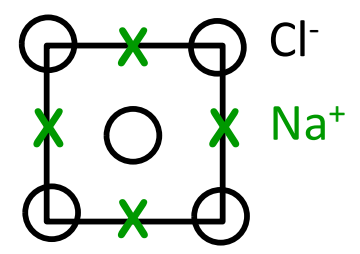
$\bigcirc$ PtCl_6^{2-}

8 K^+ & 4 PtCl_6^{2-}

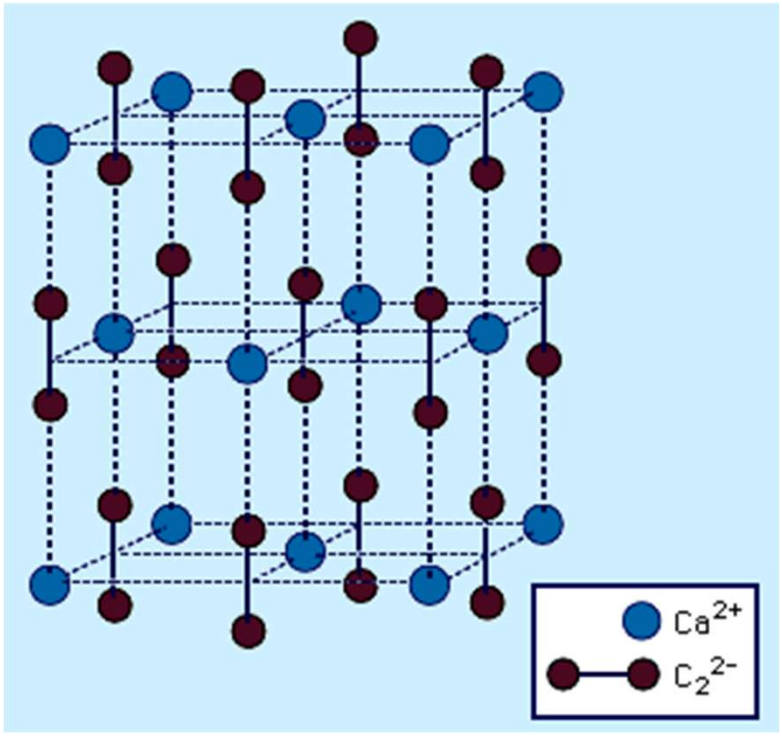
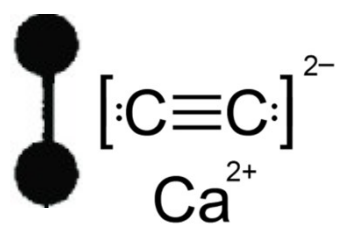
Complicated NaCl-type Structure



similar to



CdO_2 , PbSb_2 , PtSb_2



Cubic $\rightarrow$ Tetragonal

$$a' = \frac{a}{\sqrt{2}}$$

$$c \neq a$$

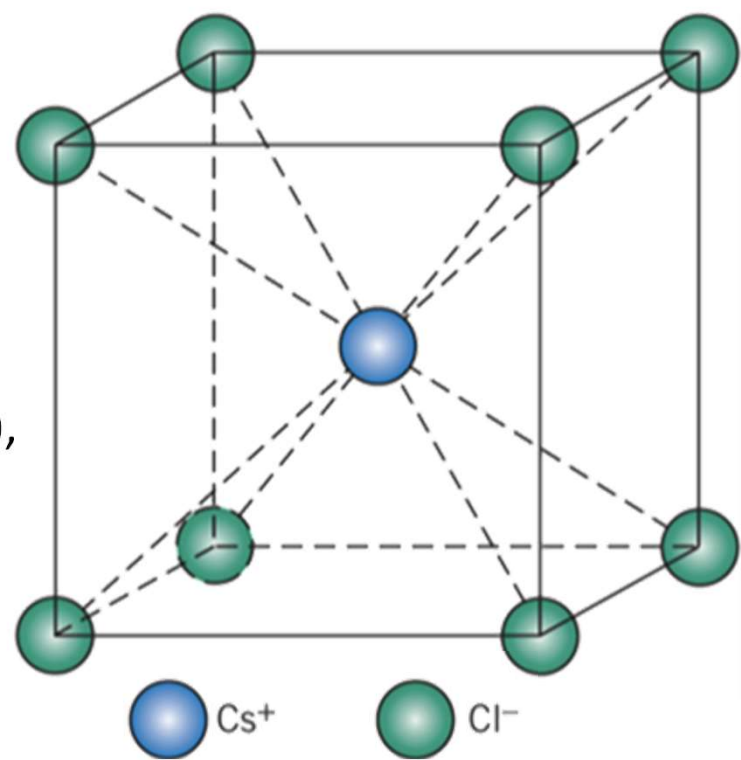
$$Ca \rightarrow 000 \quad 0\frac{1}{2}\frac{1}{2} \quad \frac{1}{2}0\frac{1}{2} \quad \frac{1}{2}\frac{1}{2}0$$

$$C \rightarrow 00z \quad 00\bar{z} \quad \frac{1}{2}\frac{1}{2}\frac{1}{2} + z \quad \frac{1}{2}\frac{1}{2}\frac{1}{2} - z$$

CaO_2 , BaO_2 , MoSi_2

Other Cubic Structures:

CsCl



β -brass (CuZn),
CsBr, CsI, AlN

A	B
000	$\frac{1}{2} \frac{1}{2} \frac{1}{2}$

In BCC, coordination is 8
In simple cube, coordination is 6

BCC lattice requires lattice points at the corners and at the center, both having the same motif

Here motif has changed!

Condition to get this structure:

(CN is 8) $\frac{R_A}{R_B} = \text{large}, > 0.73 \rightarrow \text{CsCl structure}$

same type of atom, CN is 6
different type of atoms, CN is 12

(CN is 6 or 12) $\frac{R_A}{R_B} = 0.41 - 0.73 \rightarrow \text{NaCl structure}$

TiO₂ (Rutile)

TiO₂ has 3 forms: Anatase, Brookite and Rutile

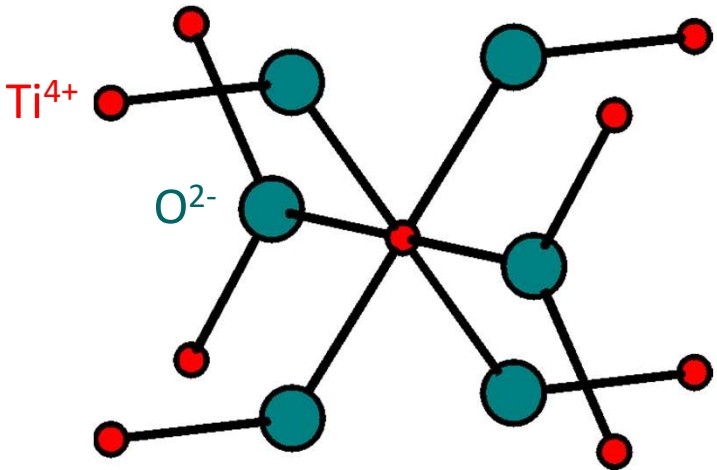
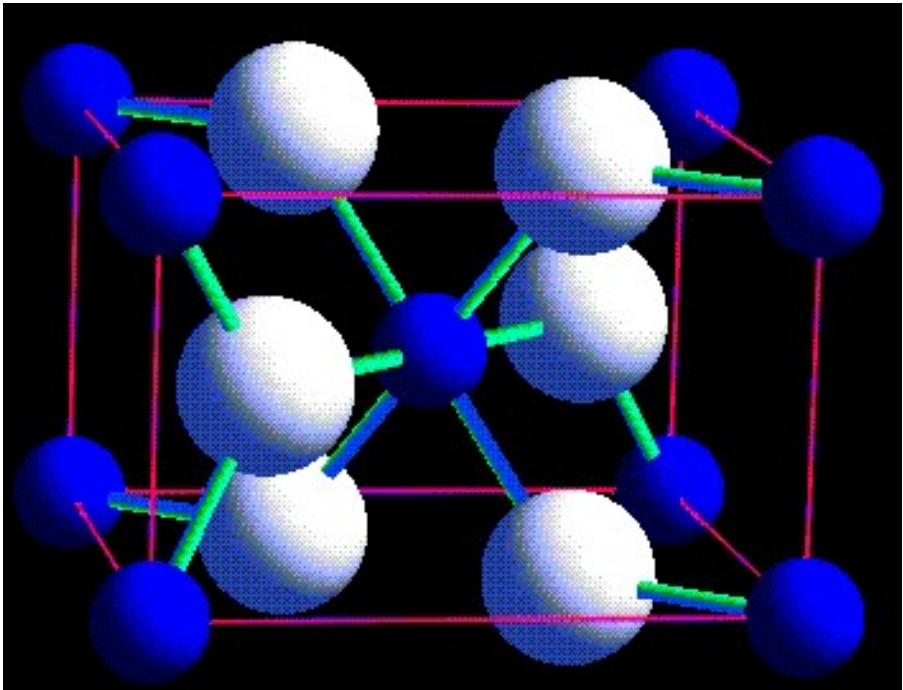
Rutile structure → radius ratio = 0.41-0.73

Each Ti⁴⁺ is octahedrally surrounded by 6 O²⁻ ions in a planar triangular arrangement

Coordination is 6 : 3

Rutile structure is not close packed. The unit cell is not a cube, one of the axes is 30% shorter than the other two.

The structure: **distorted BCC lattice of Ti⁴⁺ ions**



$$Ti \rightarrow 000 \quad \frac{1}{2} \frac{1}{2} \frac{1}{2}$$

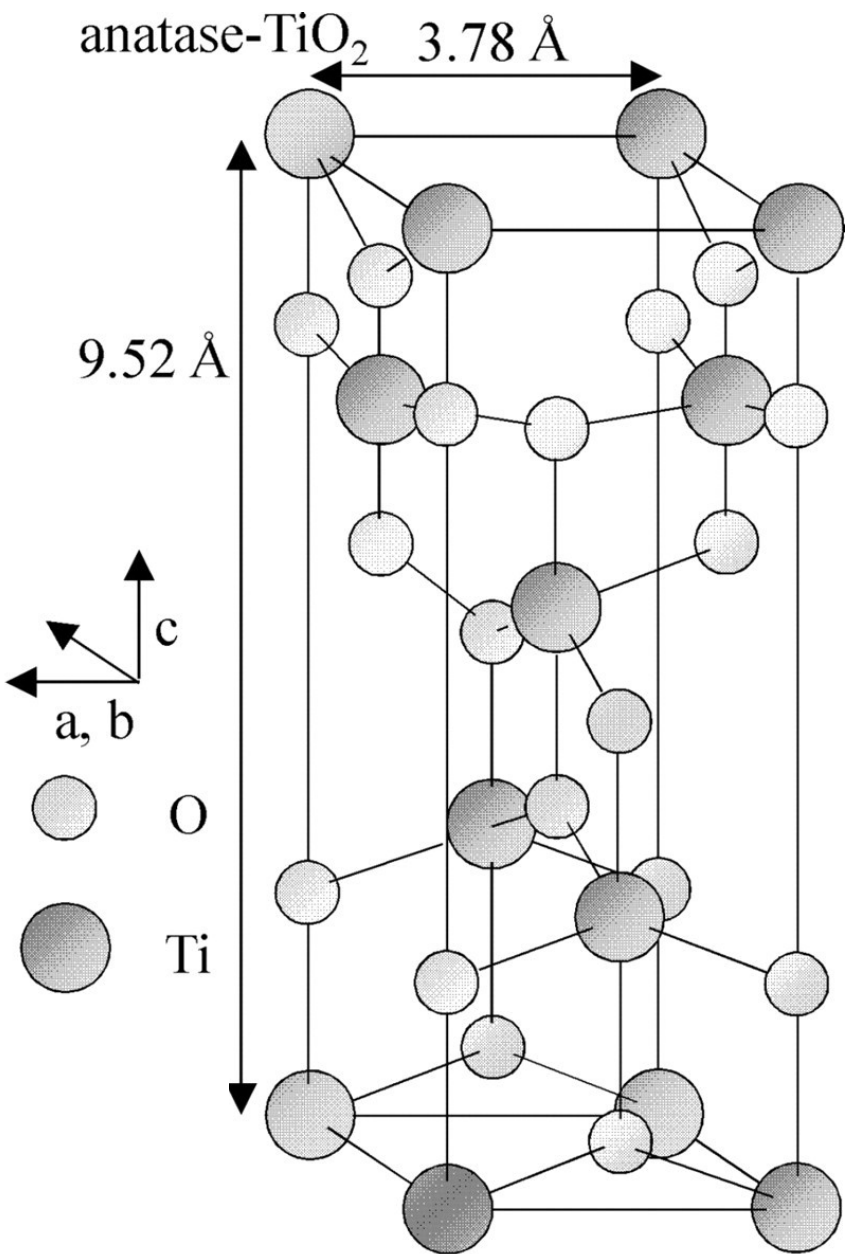
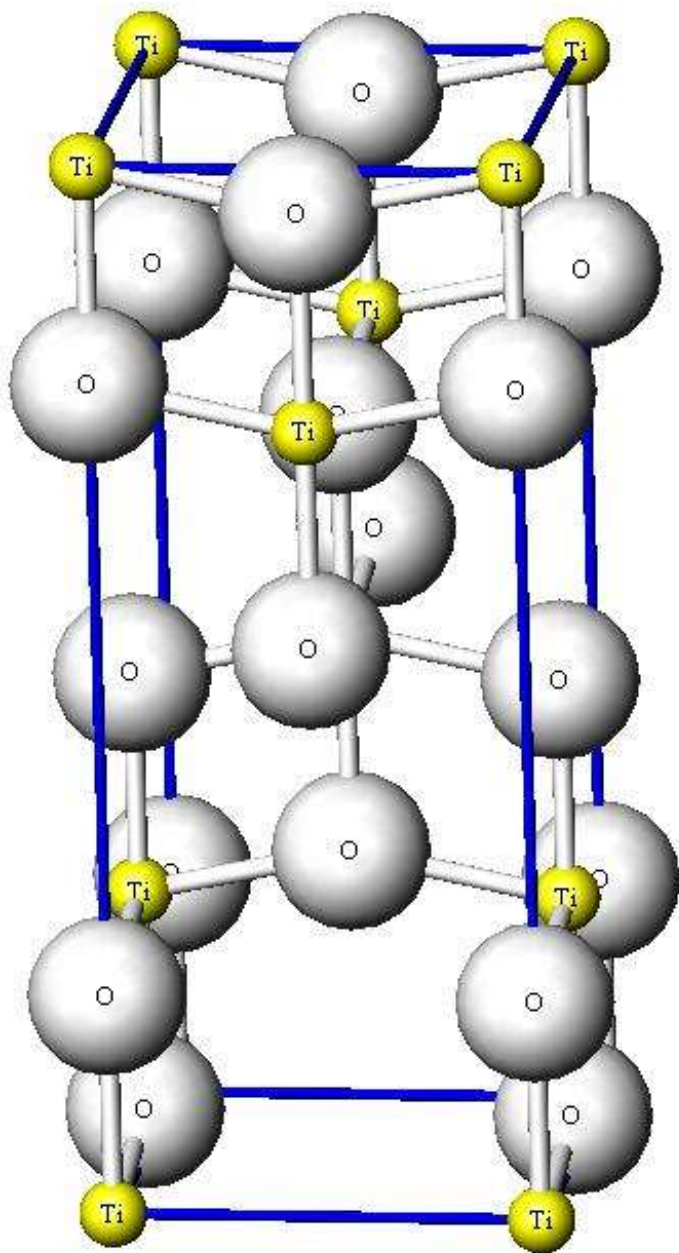
$$O \rightarrow xx0 \quad \bar{x}\bar{x}0 \quad \frac{1}{2} + x, \frac{1}{2} - x, \frac{1}{2} \quad \frac{1}{2} - x, \frac{1}{2} + x, \frac{1}{2}$$

$$\frac{1}{4} \frac{1}{4} 0 \quad \frac{3}{4} \frac{3}{4} 0 \quad \frac{3}{4} \frac{1}{4} \frac{1}{2} \quad \frac{1}{4} \frac{3}{4} \frac{1}{2} \quad (\text{putting the value of } x = \frac{1}{4})$$

Actual value: $x = 0.3$

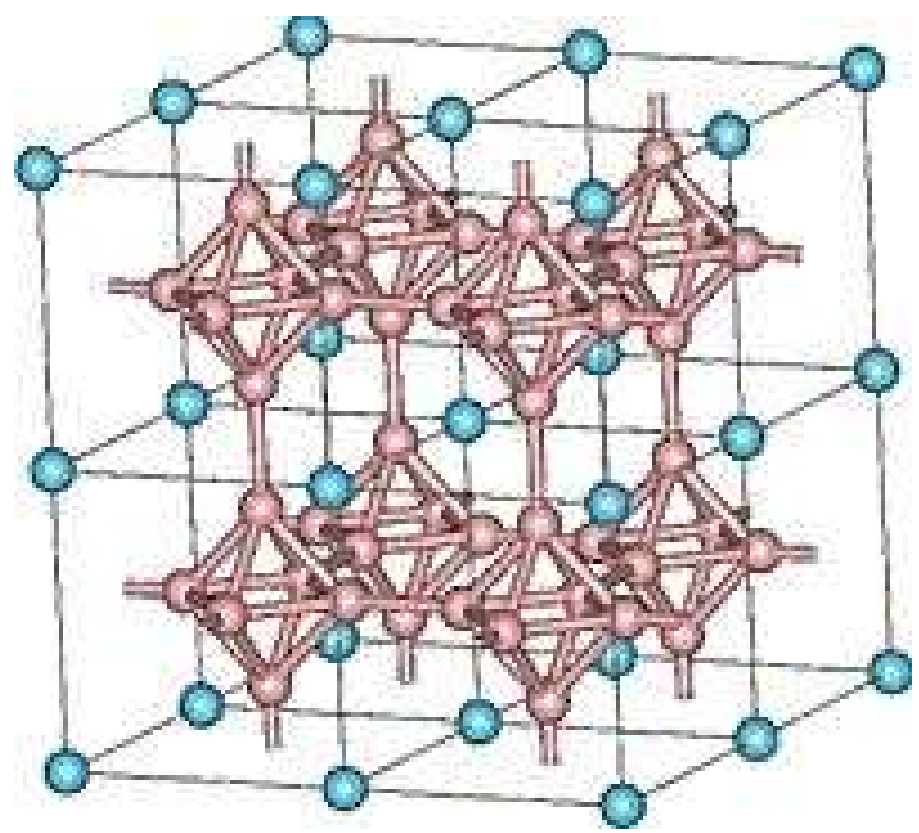
TiO₂ (Anatase)

For your knowledge only



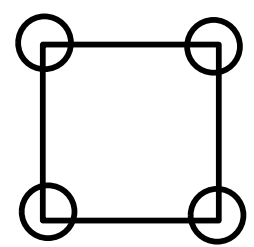
Hexaborides (CaB₆, LaB₆ etc.)

[excellent electron sources, generate large quantity of electrons when heated]

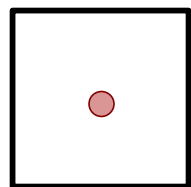


$Ca \rightarrow 000$

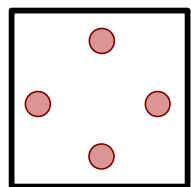
$B \rightarrow \begin{smallmatrix} 1 & 1 & 1 \\ 2 & 2 & 4 \end{smallmatrix} \quad \begin{smallmatrix} 1 & 1 & 3 \\ 2 & 2 & 4 \end{smallmatrix} \quad \begin{smallmatrix} 1 & 1 & 1 \\ 2 & 4 & 2 \end{smallmatrix} \quad \begin{smallmatrix} 1 & 1 & 1 \\ 4 & 2 & 2 \end{smallmatrix} \quad \begin{smallmatrix} 1 & 3 & 1 \\ 2 & 4 & 2 \end{smallmatrix} \quad \begin{smallmatrix} 3 & 1 & 1 \\ 4 & 2 & 2 \end{smallmatrix}$



$Z = 0, 1$



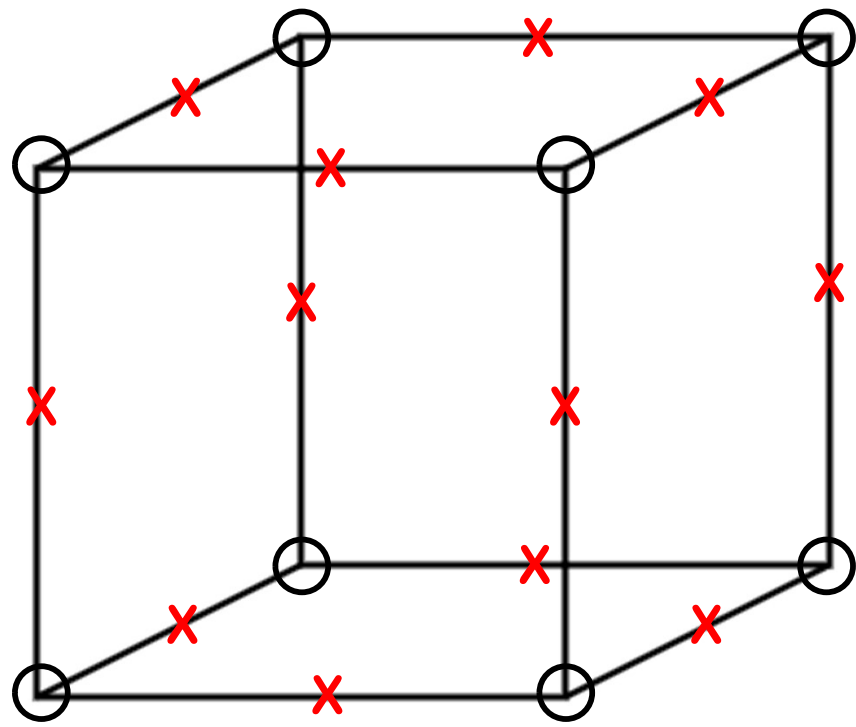
$Z = \frac{1}{4}, \frac{3}{4}$



$Z = \frac{1}{2}$

ReO₃

[BX₃ type]



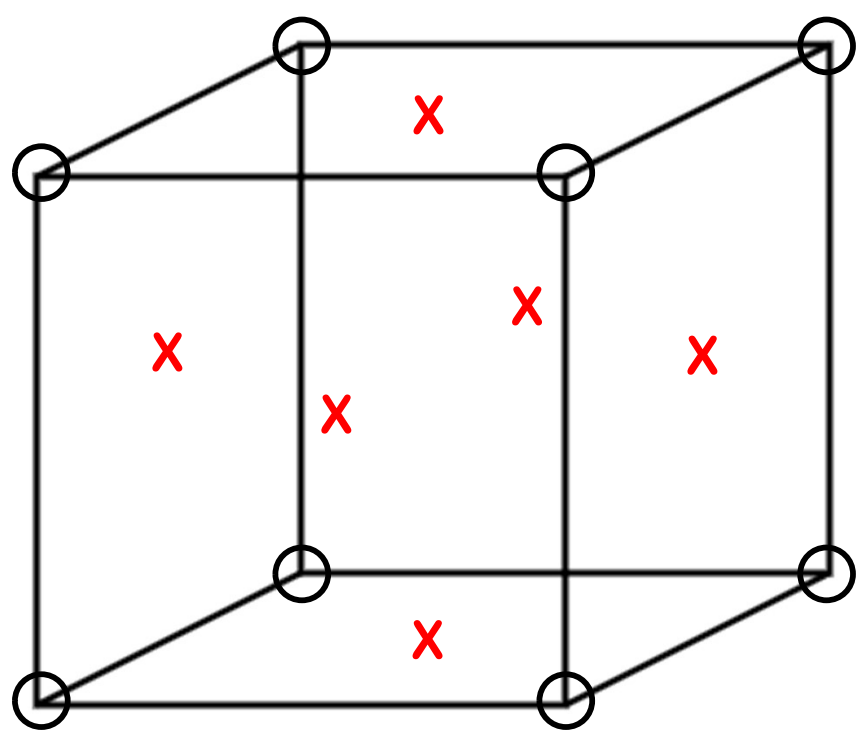
○ *Re* → 000

× *O* → $\frac{1}{2}00$ $0\frac{1}{2}0$ $00\frac{1}{2}$

Effective number of oxygen is 3

Cu₃Au

[AB₃ type]

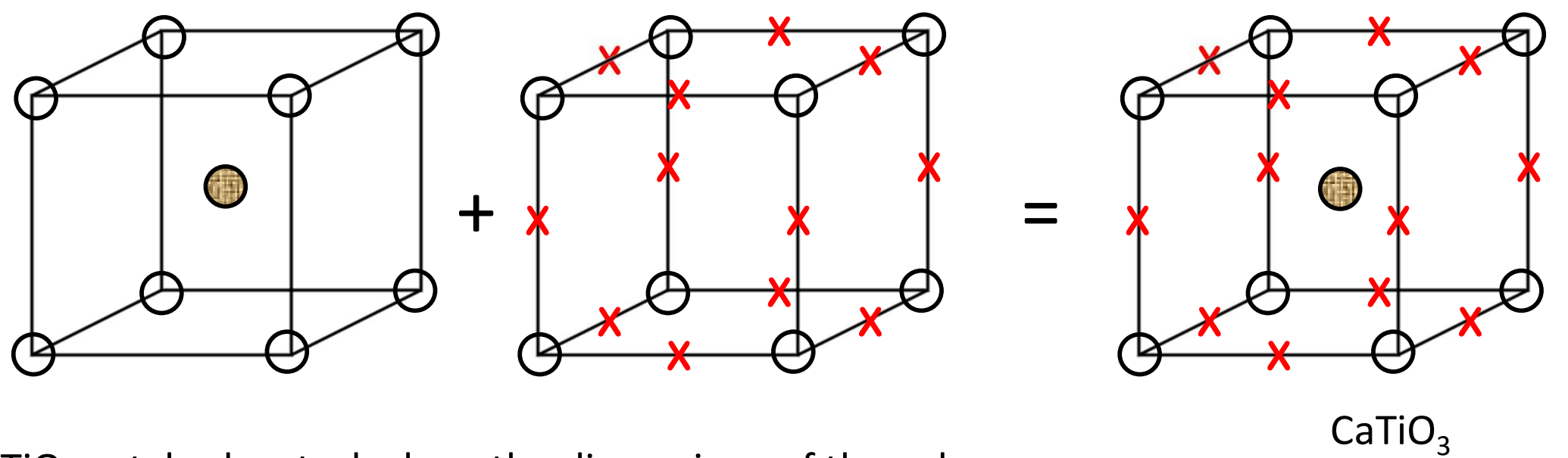


- Au → 000
- × Cu → $\frac{1}{2}\frac{1}{2}0$ $0\frac{1}{2}\frac{1}{2}$ $\frac{1}{2}0\frac{1}{2}$

Putting another atom in each face center

CsCl + ReO₃

AB + BX₃



TiO₆ octahedra stack along the dimensions of the cube

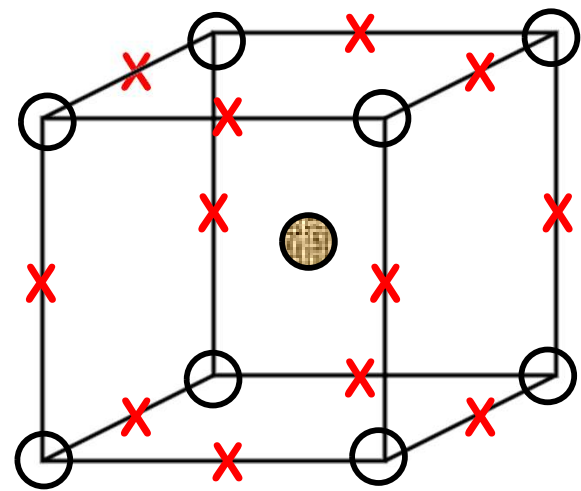
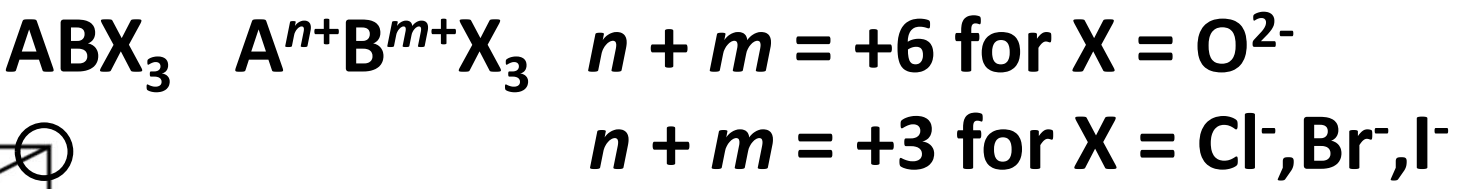
- CN of Ti → 6 (oxygen)
- CN of Ca → 12 (oxygen) [CaO₁₂] dodecahedra
- Oxygen occupy the octahedral voids

$Ti \rightarrow 000$

$Ca \rightarrow \frac{1}{2} \frac{1}{2} \frac{1}{2}$

$O \rightarrow \frac{1}{2} 00 \quad 0 \frac{1}{2} 0 \quad 00 \frac{1}{2}$

Perovskite Structure



- CaTiO₃
- SrTiO₃
- BaTiO₃
- LaCoO₃
- KNbO₃

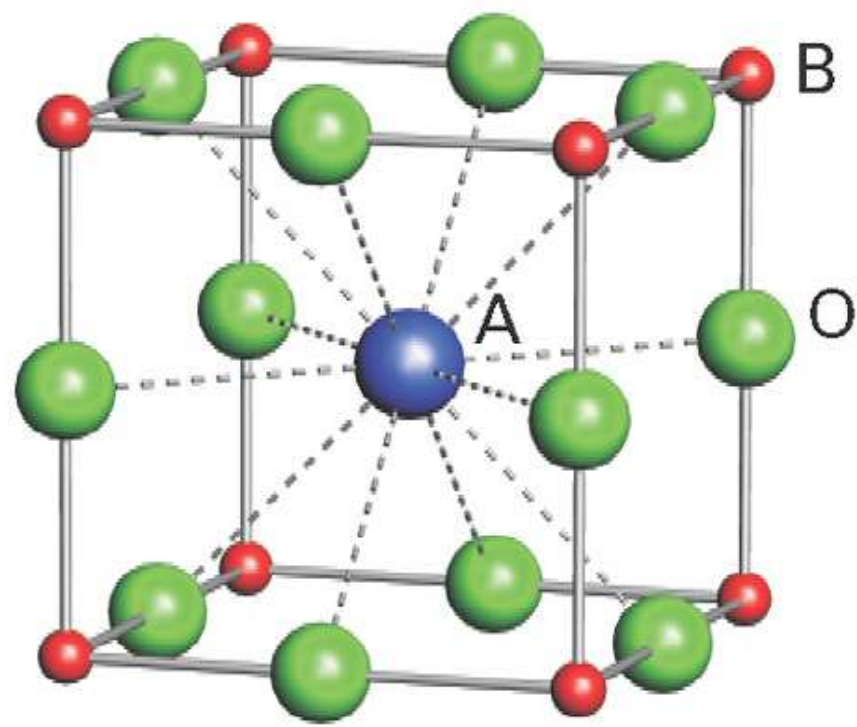
- CH₃NH₃PbI₃
- CsPbBr₃

Goldsmith Tolerance Factor (*t*)

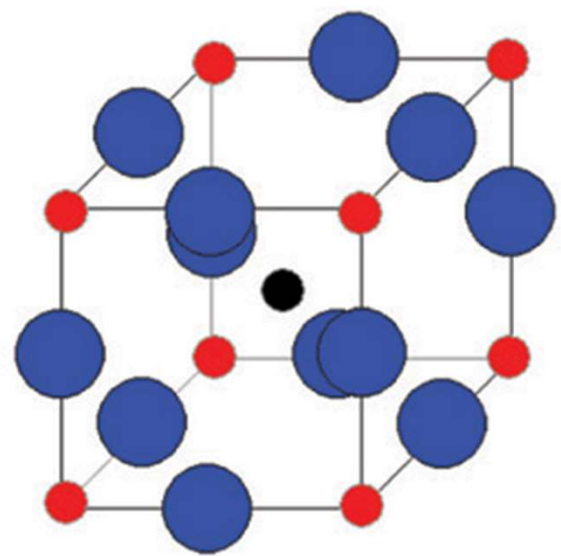
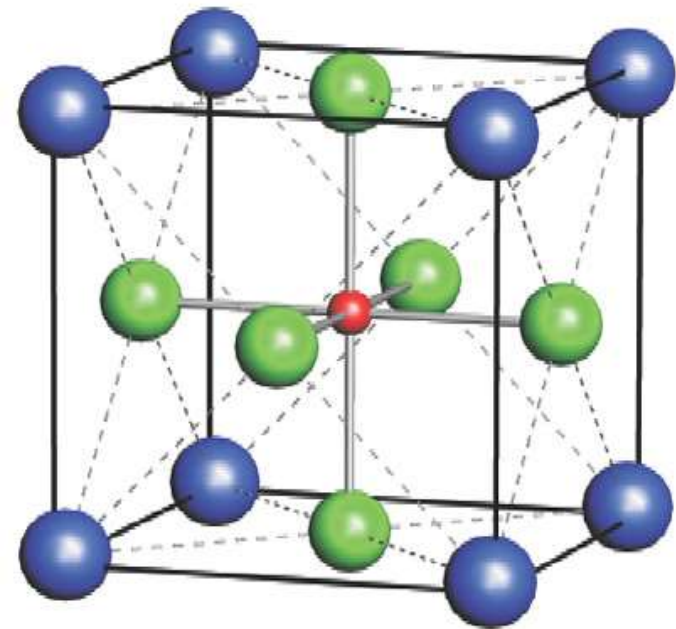
$$t = \frac{r_A + r_X}{\sqrt{2} (r_B + r_X)}$$

$t = 1$ (ideal perovskite)
 $\approx \pm 15 \%$

Perovskite Structure



This is the other way round

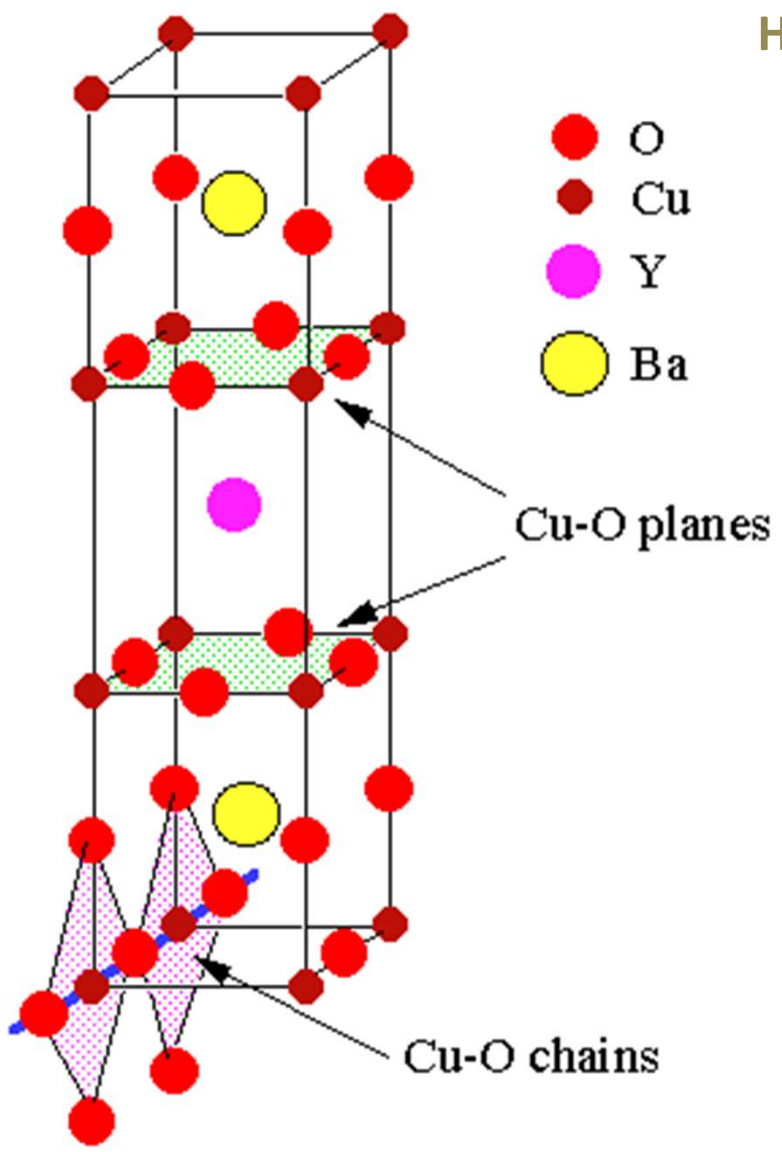


- Barium atom
- Titanium atom
- Oxygen atom

Related Structure

$\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$

High T_c superconductor



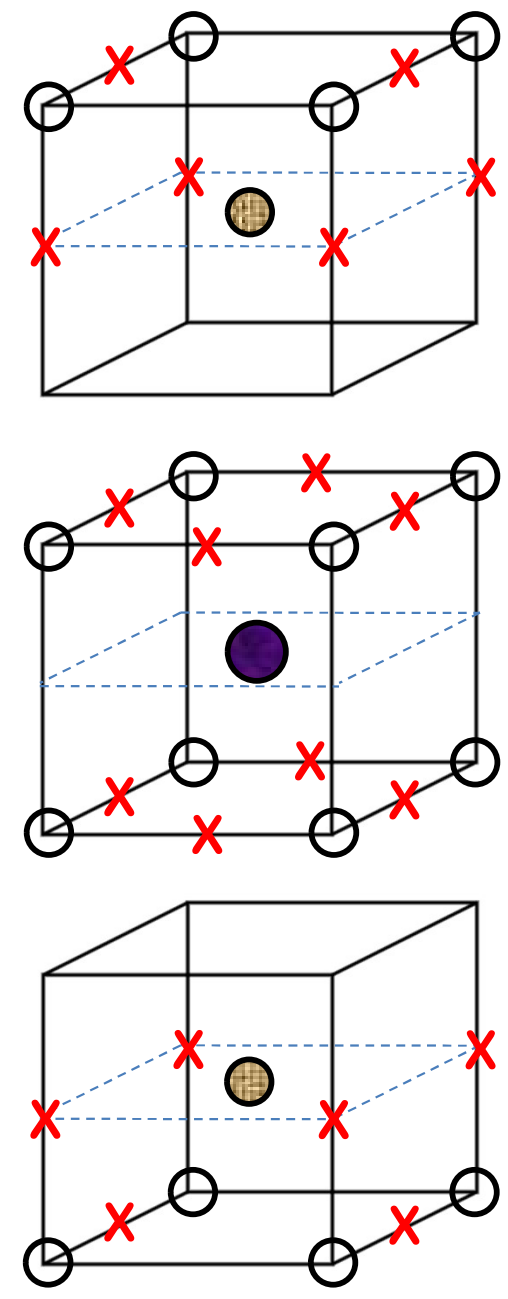
13 (+)charges
14 (-) charges

If δ is 0.5, structure will be neutral with 6.5 O
(Heat in vacuum)

If $\delta < 0.65$, orthorhombic
If $\delta = 1$, tetragonal

Oxygen non-stoichiometry

- $\text{Cu}^{2+}/\text{Cu}^{3+}$
- Ba^{2+}
- Y^{3+}
- × O^{2-}



Spinel (AB₂O₄)

Aluminates	Ferrites	Titanates	
MAI ₂ O ₄	MFe ₂ O ₄	MTi ₂ O ₄	M = Fe, Cu, Zn, Mn, Ni

Spinel structure consists of 8 blocks: 8[AB₂O₄] → 64 tetrahedral voids
32 octahedral voids

Normal spinel (e.g. ZnFe₂O₄, MgAl₂O₄)
1/8 th occupied by A²⁺ → 8
1/2 occupied by B³⁺ → 16

Inverse spinel (e.g. Fe₃O₄, MgFe₂O₄)
1/2 of B³⁺ occupy td voids → 8 sites
1/2 of B³⁺ occupy oh voids → 8 sites
A²⁺ occupy oh voids → 8 sites

Arrangement of the two types of cations that generate the most CFSE will be most stable
(PRACTICE)

When crystal field stabilization effect is NIL.

MgAl₂O₄ has a normal spinel structure. How?

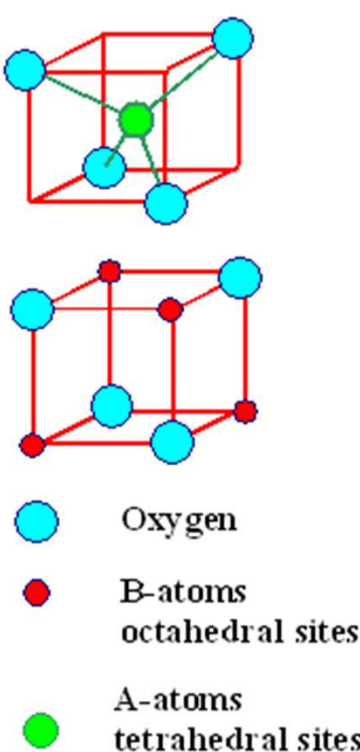
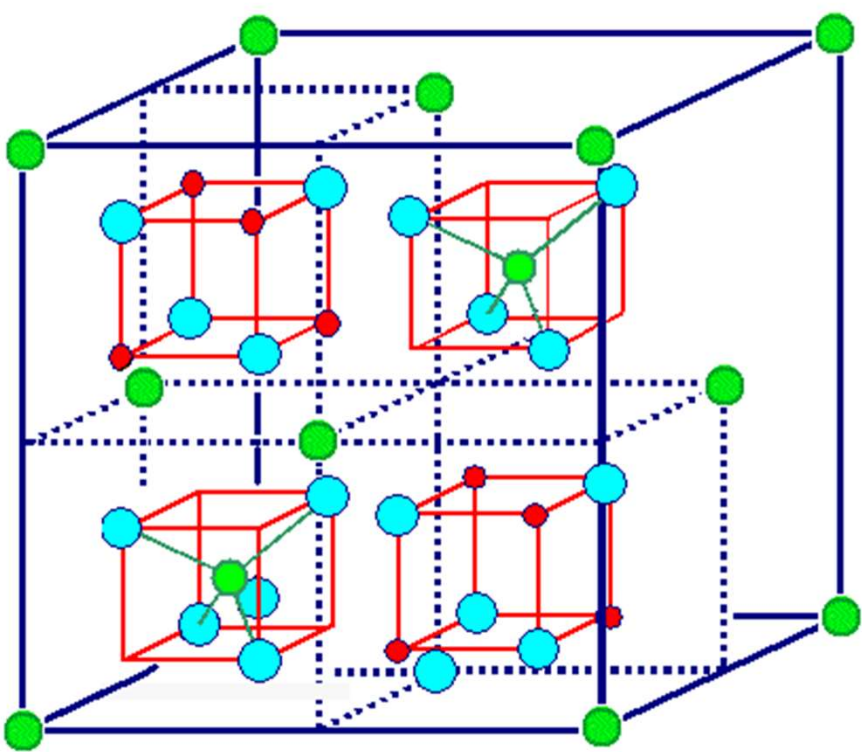
From **lattice energy** consideration, cation with higher charge provides greater lattice energy when placed in octahedral environment than the tetrahedral sites. So Al³⁺ will occupy oh sites.

Degree of Inversion λ (= 0 for normal, 0.5 for inverse)

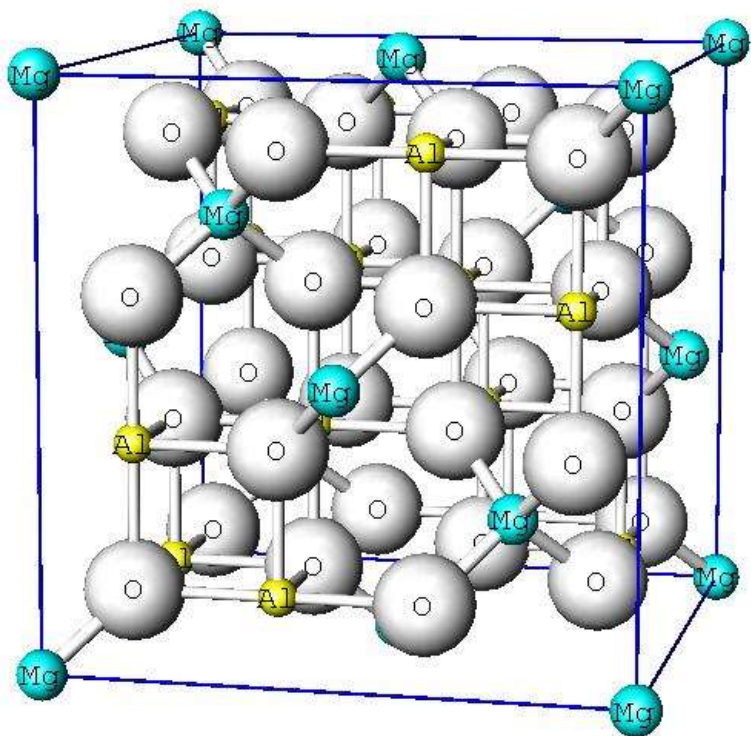
Lattice energy
$$U = \frac{N_A M z^+ z^- e^2}{4\pi\epsilon_0 r} \left(1 - \frac{1}{n}\right)$$

M = Madelung constant (crystal geometry)
ε₀ = permittivity of free space
r = distance to closest ion, n = Born exponent (5-12)

Spinel (AB₂O₄)

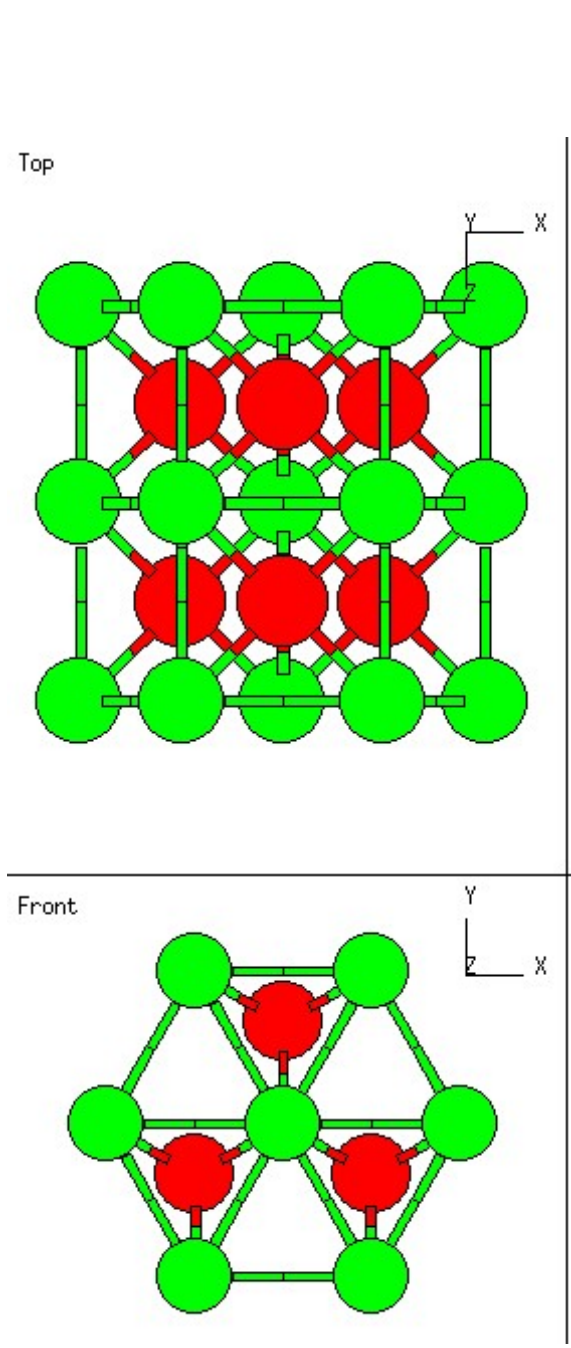


AB₂O₄ spinel The red cubes are also contained in the back half of the unit cell



E.J.W. Verwey and E.L. Heilmann , J. Chem. Phys. 15 (4) , pp. 174-180 (1947) ; Fig. 1 , p. 174 .
A unit cell with eight AB₂X₄ formula units is shown .

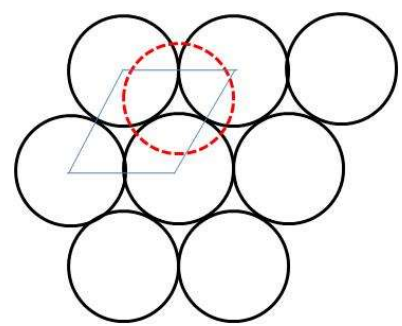
Tungsten Carbide



Hexagonal lattice is replaced by 2 atoms at:

$$\begin{matrix} 0 & 0 & 0 \\ \hline \frac{1}{3} & \frac{2}{3} & \frac{1}{2} \end{matrix} \text{ ABABAB}$$

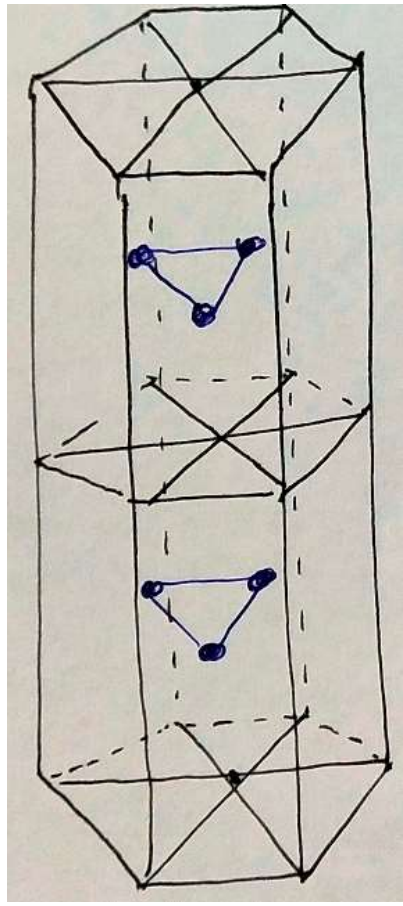
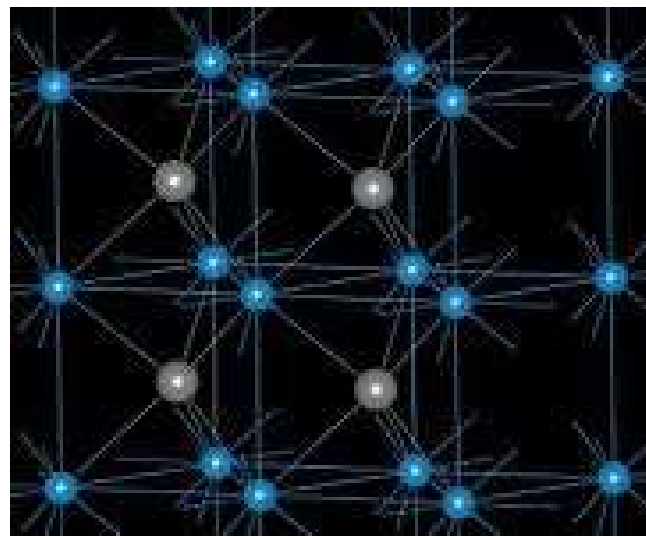
OR
$$\begin{matrix} 0 & 0 & 0 \\ \hline \frac{2}{3} & \frac{1}{3} & \frac{1}{2} \end{matrix} \text{ ACACAC}$$



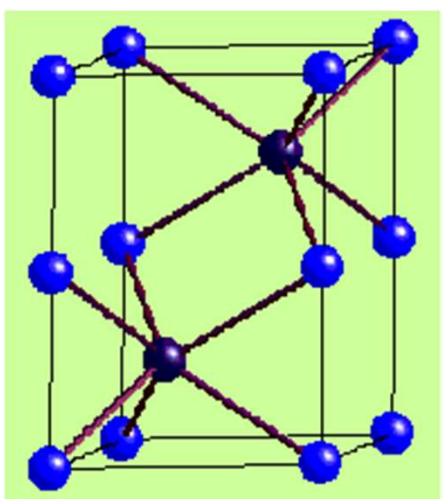
$W \rightarrow 000$

$\alpha\beta\alpha\beta\alpha\beta$

$C \rightarrow \frac{2}{3}\frac{1}{3}\frac{1}{2} \text{ or } \frac{1}{3}\frac{2}{3}\frac{1}{2}$

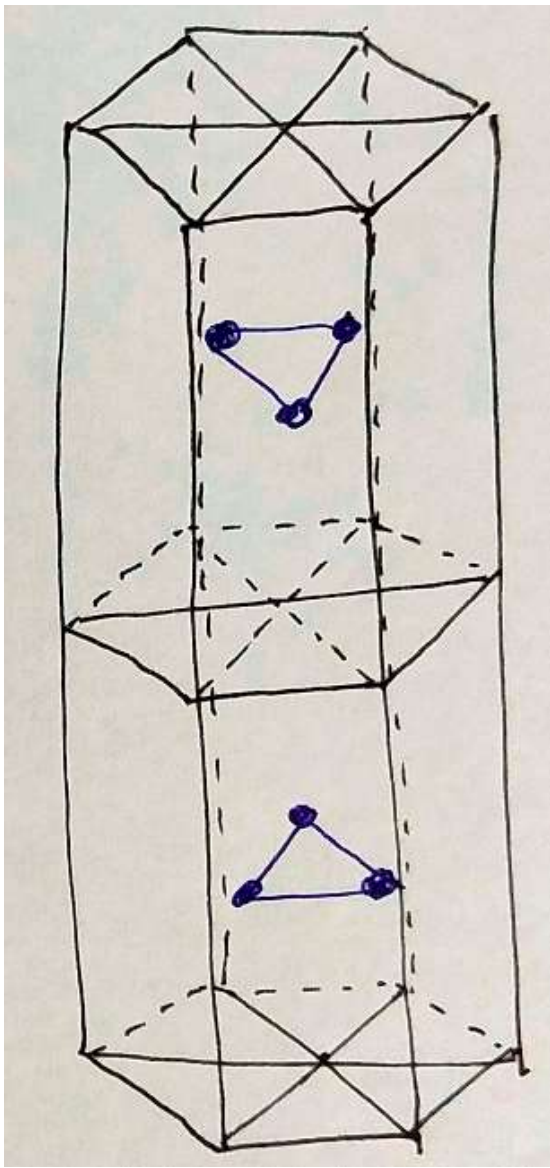
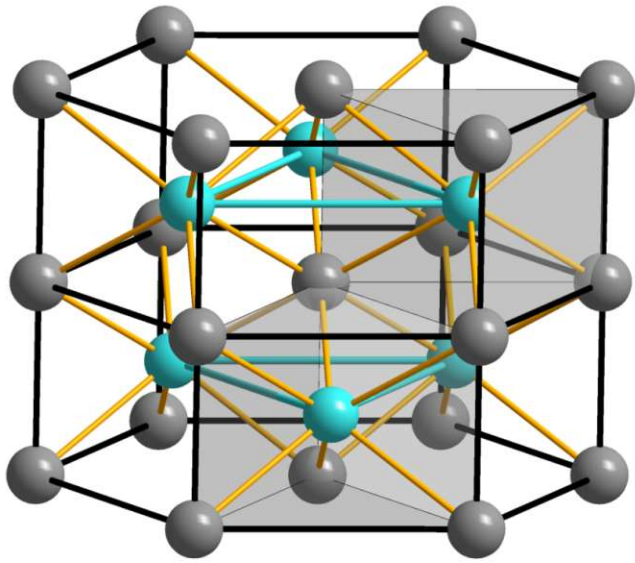


Nickel Arsenide

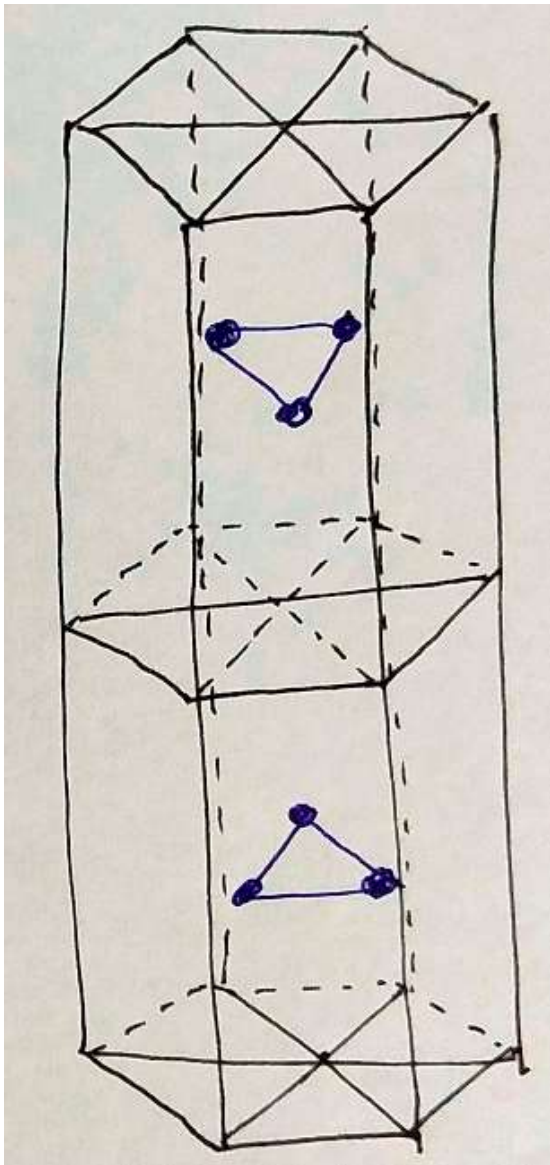
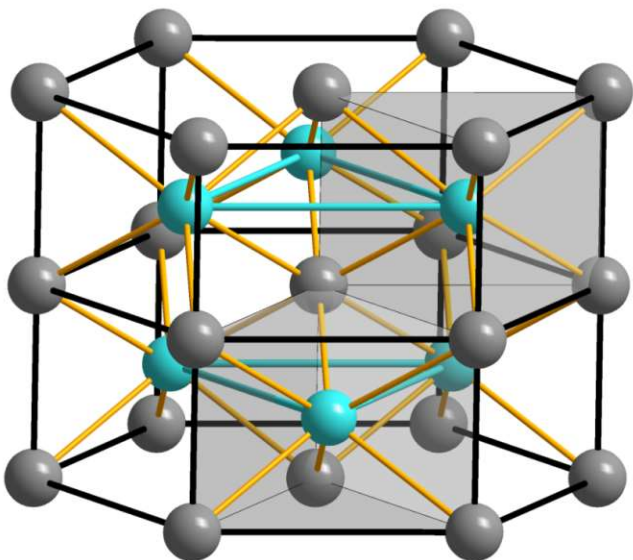


● Ni
● As

$$Ni \rightarrow 000 \quad 00\frac{1}{2}$$
$$As \rightarrow \frac{2}{3}\frac{1}{3}\frac{1}{4} \quad \frac{1}{3}\frac{2}{3}\frac{3}{4}$$



HCP of Ni with As in all Octahedral holes
2NiAs in unit cell (Cell size doubled)

$$As \rightarrow \frac{2}{3} \frac{1}{3} \frac{1}{4} \quad \frac{1}{3} \frac{2}{3} \frac{3}{4}$$


HCP of As with Ni in all Octahedral holes
2NiAs in unit cell (Cell size doubled)
Coordination: Ni 6 (octahedral) : As 6 (trigonal prismatic)

Wurtzite

2 HCP sub-lattices shifted to each other along c-axis

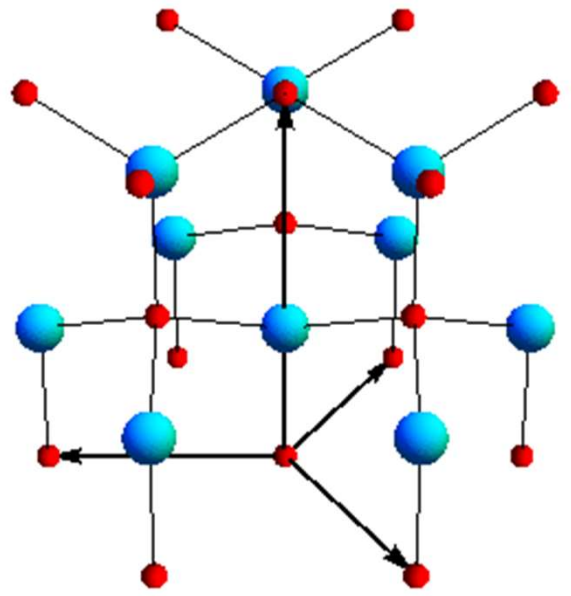
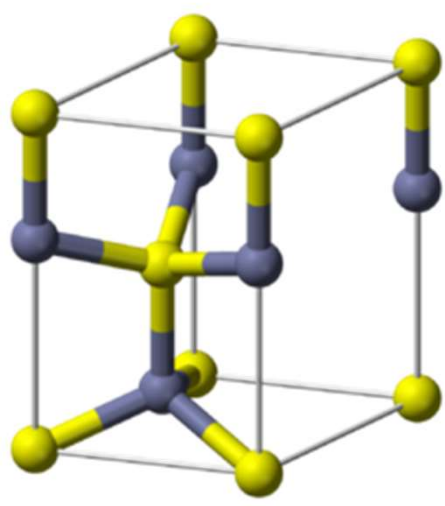
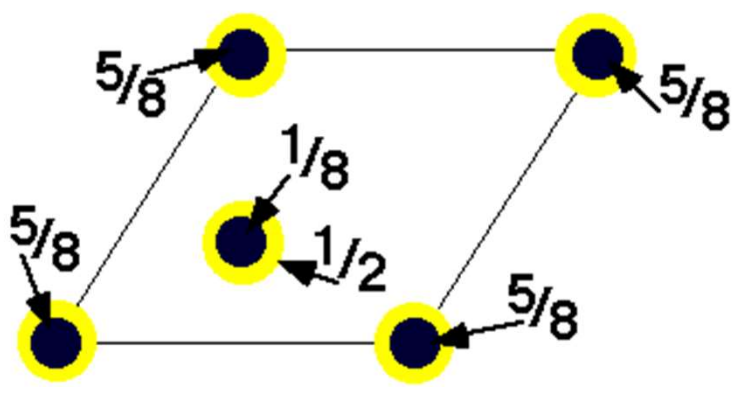
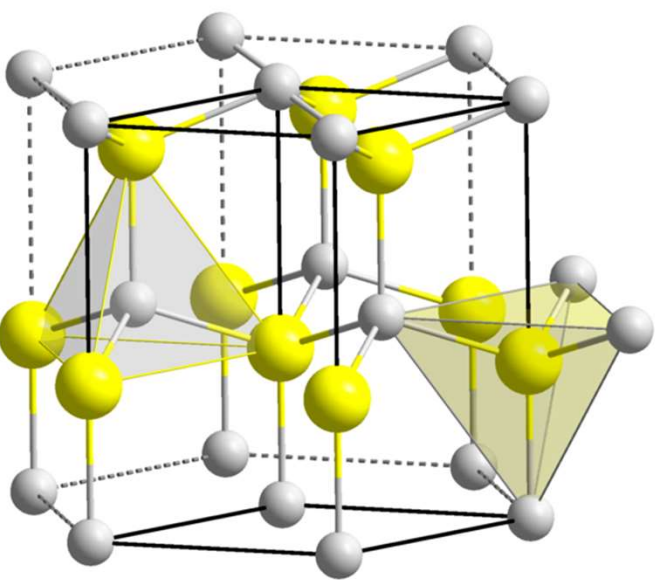
Both sub-lattices are occupied by 1 atomic species only and each atom is surrounded by 4 atoms of the other kind, which are arranged at the edges of a tetrahedron

ZnS, ZnO, ZnSe, CdS, CdSe, MnS, MnSe, AgI, AlN, GaN

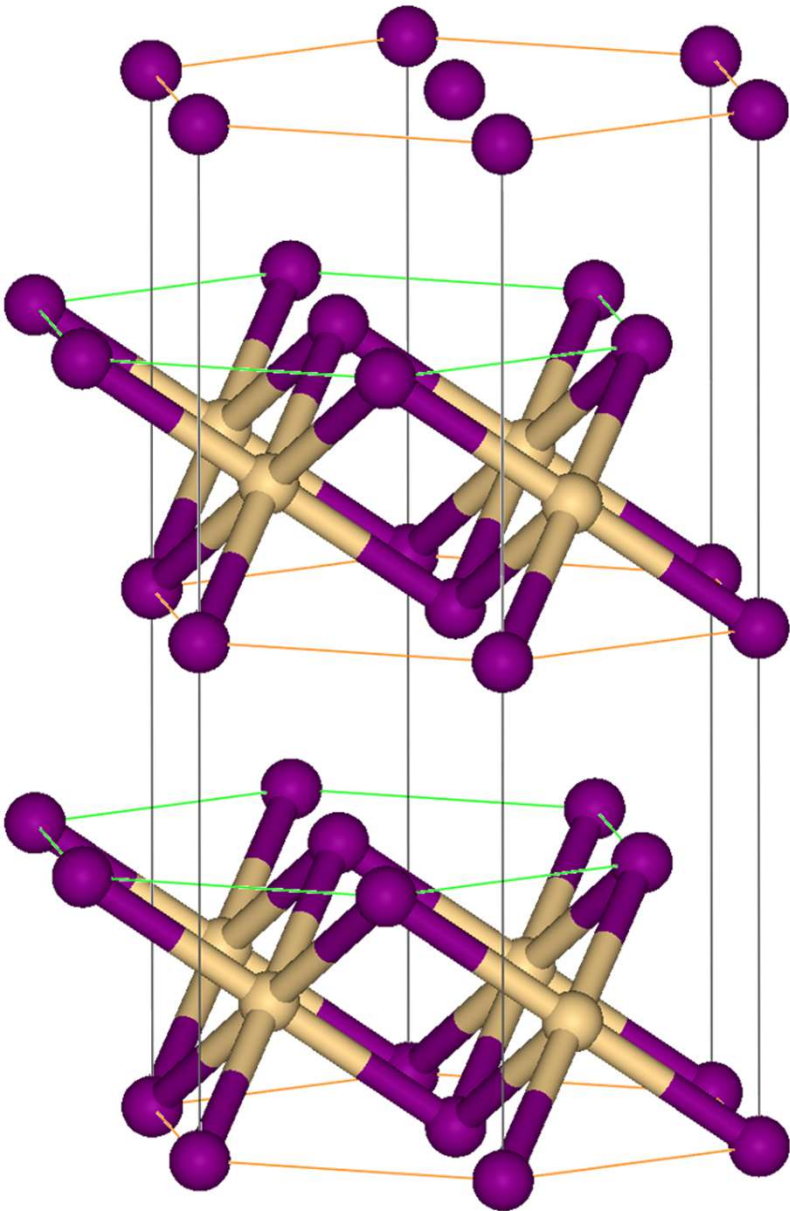
$$2S \rightarrow 000 \quad \frac{2}{3} \frac{1}{3} \frac{1}{2}$$

T₊ (or T₋) sites occupied
T₋ (or T₊), Oh empty

$$2Zn \rightarrow \frac{2}{3} \frac{1}{3} \frac{1}{8} \quad 00 \frac{5}{8}$$



Cadmium Iodide



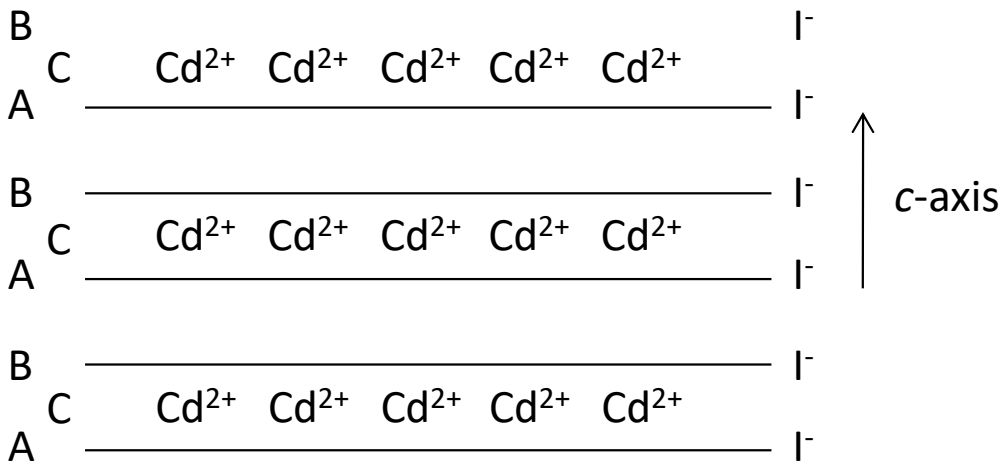
Layered Structure

I⁻ ions form HCP lattice (close packed anion array)

Cd²⁺ fill all the octahedral sites in alternate layers

$$I \rightarrow 000 \quad \frac{1}{3} \frac{2}{3} \frac{1}{2}$$

$$Cd \rightarrow \frac{2}{3} \frac{1}{3} \frac{1}{4}$$



A_2B_3 (Sesquioxide)

All anions (O^{2-}) form the HCP structure

All cations (Al^{3+}) occupy 2/3 rd of octahedral sites (1/3 rd octahedral holes empty)

All tetrahedral sites are vacant

Al_2O_3 Corundrum

$Al_{2/3}O$

Bond within the layers $\rightarrow$ strong ; Bond between the layers $\rightarrow$ weak

So anions can mobilize through this structure

$Al_2O_3 + Na^+ \rightarrow$ superionic conductors

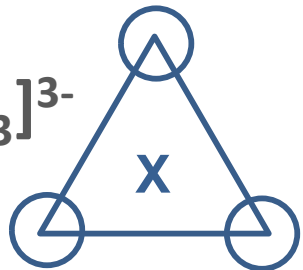
$\alpha-Al_2O_3$, Cr_2O_3 , $\alpha-Fe_2O_3$, $\alpha-Ga_2O_3$, V_2O_3

Linkage of Polyhedra Borates & Silicates

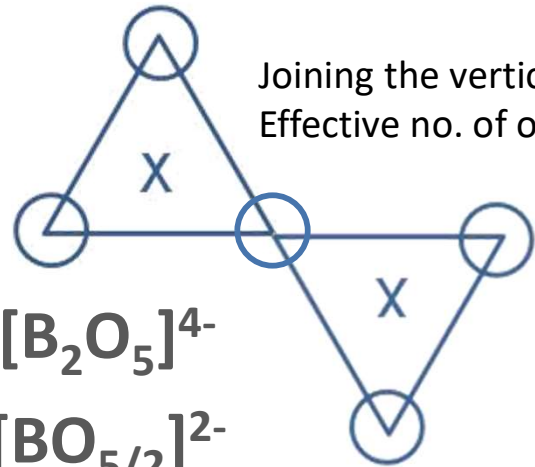
Leads to stoichiometry

Borate

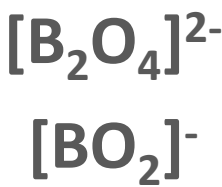
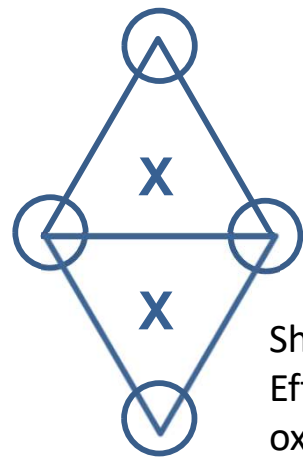
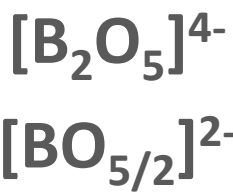
B – 3O



Trigonal planar structure
(fundamental unit
for all borates)



Joining the vertices
Effective no. of oxygen = 5

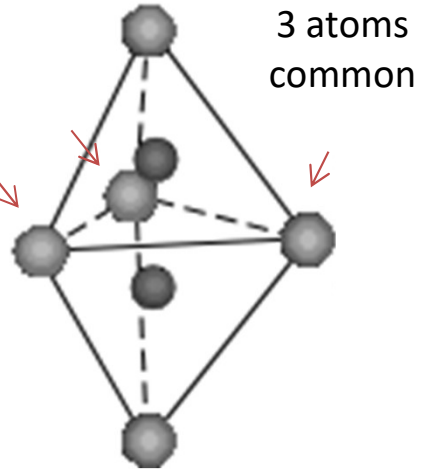
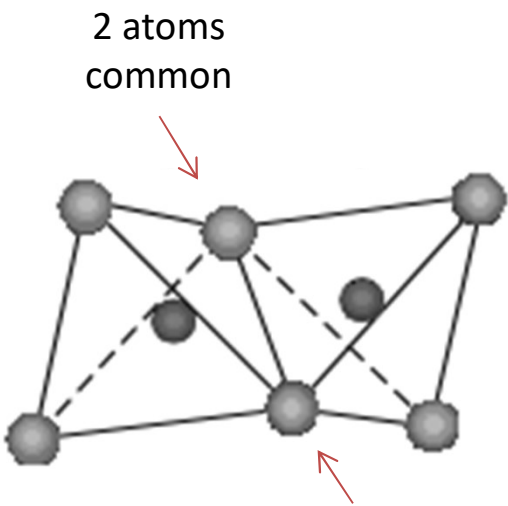
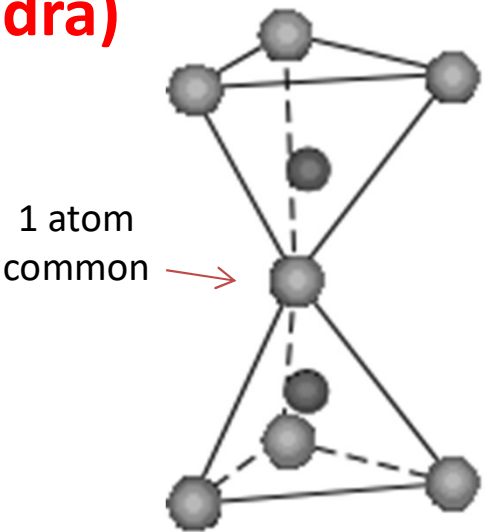


Sharing the edges
Effective no. of
oxygen = 4

Silicate (tetrahedra)

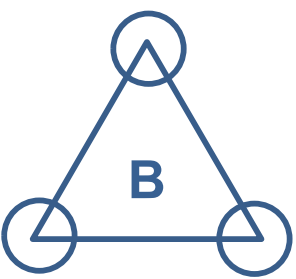
Si – 4O $[\text{SiO}_4]^{4-}$

Sharing is possible
by vertex, edge, face



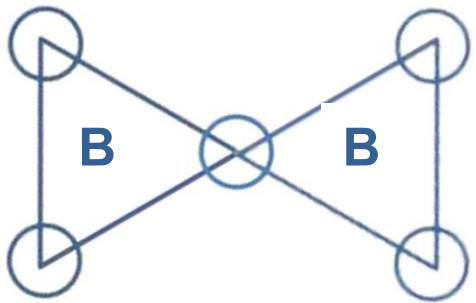
Possible sharing of oxygen in borates

0



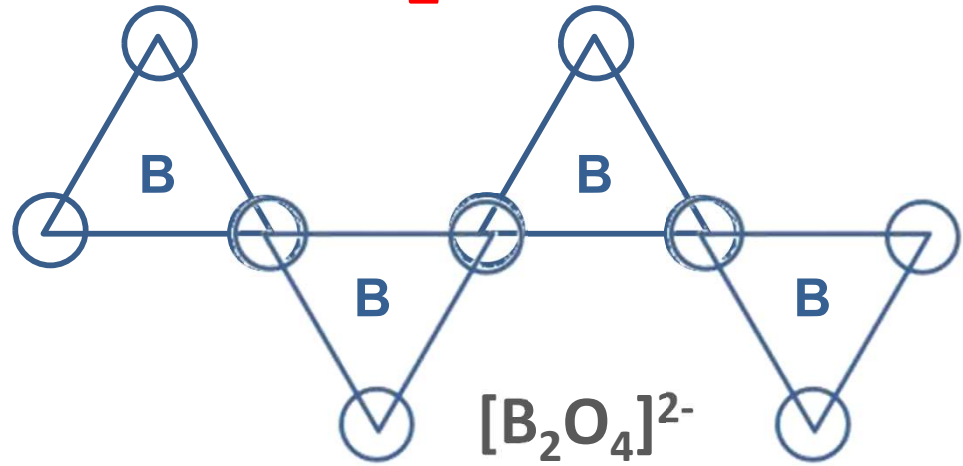
Orthoborate
 Be_2OHBO_3
Hambergite

1



$\text{Mg}_2\text{B}_2\text{O}_5$
Effective no. of O
 $= \frac{1}{2} + 2 = 5/2$

2

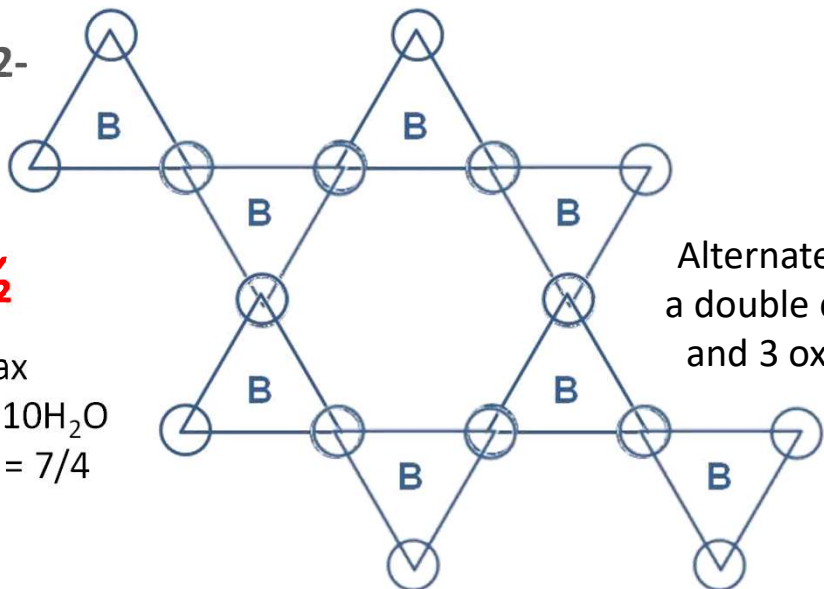


Metaborate $\text{Ca}_2\text{B}_2\text{O}_4$
Single chain



2½

Borax
 $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$
 $\frac{1}{2} + 5/4 = 7/4$



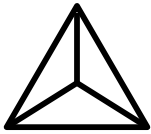
Alternate triangles in
a double chain share 2
and 3 oxygen atoms

3



Sheet form of B_2O_3
(forms a glass)

Sharing of oxygen in silicates

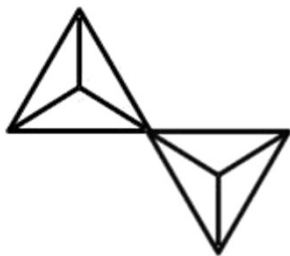


0



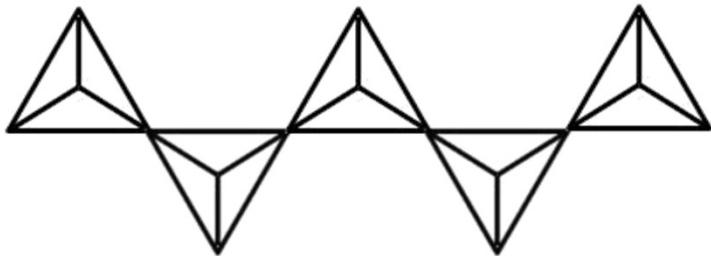
Orthosilicate
 $(\text{MgFe})_2\text{SiO}_4$
Olivine

1



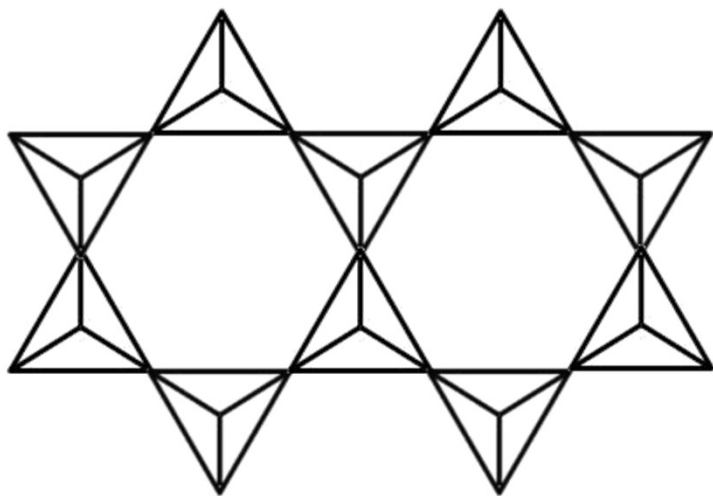
$\text{Zn}_4\text{Si}_2\text{O}_7(\text{OH})_2\text{H}_2\text{O}$
Hemimorphite

2



Single chain silicates
Layered structure
 $\text{Ba}_3\text{TiSi}_3\text{O}_9$
Benitoite

2½



Double chain silicates
 $(\text{OH})_2\text{Ca}_2\text{Mg}_5(\text{Si}_4\text{O}_{11})_2$
Tremolite

3 $[\text{Si}_2\text{O}_5]^{2-}$

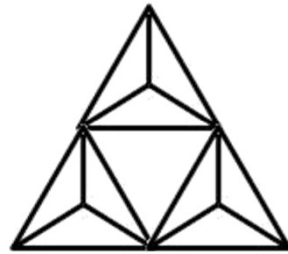
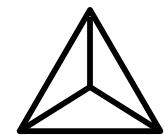
Sheets held by
van der Waals force
Cleave easily – infinite sheets
 $(\text{AlMg})_8(\text{Si}_4\text{O}_{10})_3(\text{OH})_{10} \cdot x\text{H}_2\text{O}$
Montmorillonite

4 $[\text{SiO}_2]$

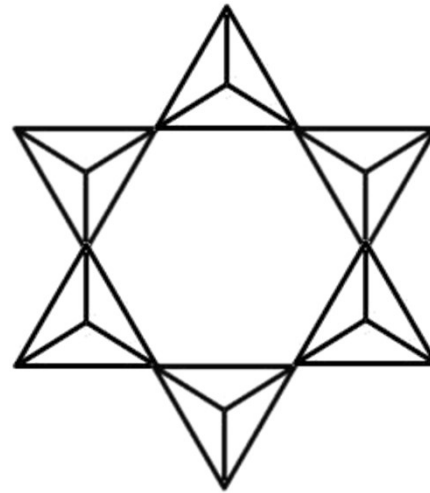
Infinite 3D Sheets
Quartz, Tridymite, Cristobalite

Sharing of oxygen in silicates

Ring Structures



3 tetrahedrons form a ring



6 tetrahedrons form a ring

Benitoite $\text{BaTiSi}_3\text{O}_9$: lots of vacancy – organic molecule packing – slow release